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AI and Economic Transition | North America

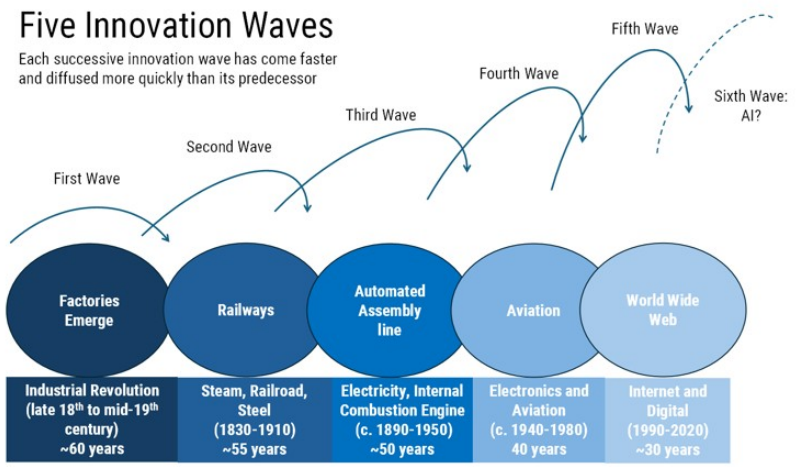
Lessons from the Five Innovation Waves That Preceded AI

The current AI wave has fueled both optimism and pessimism: it promises faster productivity and economic growth, but also raises concerns about job displacement, inequality, and social upheaval. In assessing what AI may mean for the US economy, we look to the past for guidance.

Key Takeaways

- Productivity gains are likely. AI, like prior general purpose technologies, should raise output per worker, particularly with organizational change.
- Labor displacement is transitional, not permanent. Past fears of mass technological unemployment did not materialize; employment persisted.
- Education and re-skilling will be decisive. Every prior innovation wave required expanding education, but also altering how education is provided.
- Boom-bust cycles are probable. Heavy AI infrastructure investment resembles prior railroad and telecom buildouts; financial excess and volatility are likely.
- Inequality risks are elevated. AI's effects may intensify concentration when income and wealth inequality sit at 125-year highs.

The Five Innovation Waves in the US over the past 250 Years



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MORGAN STANLEY & CO. LLC

Michael T Gapen
Chief US Economist
Michael.Gapen@morganstanley.com +1 212 761-0571

Seth B Carpenter
Chief Global Economist
Seth.Carpenter@morganstanley.com +1 212 761-0370

Sam D Coffin
Economist
Sam.Coffin@morganstanley.com +1 212 761-4630

Diego Anzoategui
Economist
Diego.Anzoategui@morganstanley.com +1 212 761-8573

Heather Berger
Economist
Heather.Berger@morganstanley.com +1 212 761-2296

Arunima Sinha
Global Economist
Arunima.Sinha@morganstanley.com +1 212 761-4125

Lingdi Xu
Economist
Lingdi.Xu@morganstanley.com +1 212 761-2957

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inequality. But they ultimately raise productivity, expand output, and fundamentally reshape the economy. While technological revolutions invariably bring short-term disruptions and uncertainty, they also deliver significant long-run benefits. How those benefits are distributed depends on how policymakers, businesses, and educators manage the transition.

Executive summary

Five Innovation Waves Before AI: Common Patterns and Lessons

The current AI wave has led to both optimism and pessimism: it promises faster productivity and economic growth, but also raises concerns about job displacement, inequality, and social upheaval. In assessing what AI may mean for the US economy, we look to the past for guidance. After all, as the saying goes, history may not repeat itself, but it often rhymes.

Over the past 250 years, the US has experienced five major innovation waves:

- The First Industrial Revolution (factories, steam, canals, railroads)
- Steam, Railroads, and Steel
- Electrification and the Internal Combustion Engine
- Electronics and Aviation
- The Internet and Digital Networks

Across these waves, a consistent pattern emerges between innovation and macroeconomic transformation: transformative technologies initially disrupt labor markets and financial systems, provoke political responses, and widen inequality – but they ultimately raise productivity, expand output, and fundamentally reshape the structure of the economy.

A look at prior innovation waves suggests that while technological revolutions invariably bring short-term disruptions and uncertainty, they also deliver significant long-run benefits. However, how those benefits are distributed – narrowly or broadly – depends on how policymakers, businesses, and educators manage the transition.

Common Themes Across the Five Innovation Waves

Across 250 years of American economic history, each innovation wave fundamentally reorganized the structure of the economy. The first two waves pulled the country out of its agrarian roots and into an industrially integrated economy, connecting regional markets through canals and railroads. Electrification and motorization then enabled mass production and national consumer markets. Electronics and aviation shifted the economy toward services and science-based industry. The digital revolution completed the transition from an agrarian to industrial to knowledge-based economy built on networks and intangible capital. Each wave did more than introduce new tools: it rewired how the economy functioned, where people lived, how firms organized production, and how value was created.

These transformations were powered by extraordinary surges in capital spending. Innovation waves are never incremental; they have been correlated with investment booms. Canal construction absorbed roughly 1% of GDP at its peak, while railroads commanded up to 2–3% of GDP annually and more than 10% of gross capital formation. Electrification required nationwide utility buildouts, federal mandates, and industrial reconfiguration. Innovation often requires greater public-private partnership: the Cold War

era saw federal R&D quadruple as a share of GDP. The digital age redirected investment from factories and structures toward software, computing equipment, and intellectual property. Every wave demanded heavy, front-loaded capital deepening, with early waves focusing on physical capital and structures and later waves on computers, peripherals, and software. The technologies were capital-intensive, network-based, and financed by optimistic projections of permanent growth.

Labor markets were never spared, though innovation largely complemented employment. As we have previously written about, each innovation wave displaced existing occupations and reshuffled skill demand. The first industrial revolution reduced the role of artisans and expanded semi-skilled factory labor. The railroad and steel era accelerated urbanization and created the modern managerial class. Electrification further reduced farm employment and expanded clerical and professional work. Postwar electronics and aerospace strengthened technical occupations while manufacturing employment peaked and then began to decline. The digital revolution hollowed out middle-skill routine jobs, pushing employment toward high-skill professional roles and lower-wage services. Yet across all five waves, the total demand for labor did not collapse. Workers were reallocated rather than rendered obsolete. Even during severe downturns, there is no evidence that technological progress permanently raised the natural rate of unemployment. The composition of work changed; work itself endured.

Productivity growth is the ultimate signature of these waves. Early diffusion is often slow and uneven, but once complementary investments and organizational change take hold, output per worker accelerates. The first industrial revolution generated modest but rising gains. The steam and steel era doubled productivity growth rates relative to the prior century. Electrification and motorization produced the "Great Leap" of the early 20th century, doubling output per hour in a generation. The postwar period delivered one of the strongest sustained productivity expansions in U.S. history. The internet wave produced a sharp acceleration in the late 1990s, even if gains later moderated. Across waves, productivity surges are real, but they require time, diffusion, and structural adaptation.

Macroeconomic volatility and boom-bust cycles are not accidents of innovation, they are necessary companions. The canal boom collapsed in 1837. Railroad overbuilding fueled the panics of 1873 and 1893. The electrification era encompassed the Great Depression. The late 1990s tech boom ended in a dramatic equity collapse. Each wave exhibited a similar pattern: exuberant capital formation, speculative financing, rising leverage, and eventual retrenchment. Financial markets both accelerate technological diffusion and amplify the cycle when expectations adjust. Innovation waves have repeatedly contained at least one major boom-bust episode. As Milton Friedman might say, "there is no free lunch."

Politics reshapes itself around these disruptions. Industrialization produced early labor organization and Progressive Era reforms. The Gilded Age's concentration of wealth triggered antitrust and income tax legislation. The third wave produced the New Deal and labor protections. The fourth wave's prosperity supported a mid-century social compact before later deregulation. The digital era has reopened debates about antitrust, data governance, and market concentration. Technological change consistently redistributes income and power, and political institutions eventually respond. Often slowly, sometimes dramatically.

Inequality has moved differently across waves but follows a recognizable arc. Early phases of innovation often widen disparities as capital deepening and new skill premiums benefit early adopters. The Gilded Age exemplified extreme concentration. The mid-20th century stands out as an exception: electrification and mass production coincided with the "Great Compression," supported by unions, progressive taxation, and expanding education. The digital revolution reversed that compression. Since 1980, income and wealth concentration have risen sharply, driven by returns to capital, skill-biased technical change, and public policy choices that reversed Great Compression-era policy. Innovation itself does not predetermine inequality: institutions and public policy mediate how gains are distributed.

Education has been a decisive adaptation mechanism. Each previous wave raised the return to human capital, and each time the United States expanded education in response. The common school movement followed early industrialization. Land-grant universities professionalized engineering and agricultural science. The high school movement and GI Bill democratized secondary and higher education. The digital age intensified demand for STEM skills and lifelong learning. When education expanded broadly, productivity gains translated into shared prosperity. When skill formation lagged, inequality widened.

Finally, diffusion has accelerated over time: each new innovation wave is more compressed than the last. The industrial revolution took half a century to spread. Railroads and steel diffused over several decades. Electrification and motorization required a generation of complementary investment. Electronics matured over roughly forty years. The internet reached mass adoption in less than thirty. Each successive wave has traveled faster than its predecessor, reflecting better communication, deeper capital markets, and accumulated institutional experience. If this pattern holds, artificial intelligence may diffuse faster still.

Taken together, the historical record is clear. Innovation waves are disruptive, capital-intensive, and volatile. They displace workers, concentrate gains early, and provoke political backlash. But over time they raise productivity, restructure labor markets, expand output, and—when institutions adapt—improve living standards broadly. The speed of diffusion is increasing, compressing adjustment timelines. The lessons are not about whether transformation will occur, but about how society manages it.

Exhibit 1: Key Elements of Past Innovation Waves

	Timeframe and Adoption Speeds	Productivity	Investment	Labor Market
Indudstrial Revolution	Innovations were adapted over a span of ~50 years. The 1790s - 1840s saw the spread of steam power and factory-based manufacturing, with progress gradual at first.	Data are limited, but we estimate real output per person employed rose by an average of 0.84% per year between 1800 and 1850. Gains across decades were uneven.	Total public and private investment in canals are estimated at about 1.0% of GDP per year at the peak. This is equivalent to about $315bn in current dollar terms based on the BEA's advance estimate of 4Q 2025 US GDP.	From 1800 - 1850, the share of farm employment slipped from around 75% to just over 50%. The labor market shifted from a skilled workforce to low-skilled or semi-skilled workforce as new factories emerged.
Steam, Railroads, and Steel	Lasted from 1830s through early 1900s. Railrods mileage grew from ~30,000 miles in 1860 to ~192,000 miles by 1900.	By the late 19th century, labor productivity growth approached 2% annually, or about double the rate of the first industrial revolution. Again, gains were unevenly distributed.	From 1872-1881,railroad investment averaged 2.5% of GDP – about $790bn in current dollars. Financing relied heavily on bond issuance marketed domestically and in Europe.	Agriculture share of labor fell further from >50% in 1860 to ~30% by 1910. Manufacturing expanded to ~25% of the workforce by the early 1900s and new white-collar occupations emerged.
Electricity and Internal Combustion Engine	Wave dated from 1890s-1940s. Electricity generation increased more than tenfold between 1902 and 1929. Ford's Model T was first produced in 1908, and car ownership grew drastically during the 1920s.	Growth in total factor productivity averaged about 1.5 percent annually from 1909 to 1929. Real output per hour in nonfarm business doubled between 1900 and 1929.	The 1920s saw speculative booms and investment in electric holding companies and auto manufacturers, but cyclical volatility followed. Fixed costs and heavy borrowing deepened the effects of the Depression.	Agriculture share of labor fell further to about 20% by 1940. Between 1910 and 1950, the clerical workforce roughly tripled, and by the mid-20th century white-collar workers outnumbered blue-collar workers.
Electronics and Aviation	This wave is generally dated generally from the 1940s through the 1980s. Television diffused rapidly in the 1950s and 1960s.	From 1945 - 1973 real GDP growth averaged over 4% annually, with labor productivity up roughly 2.5-3% per year.	Federal government spending through research and development outlays during the war and subsequent Cold War accelerated radar, jet propulsion, nuclear engineering, and computing.	By 1970 service-providing sectors employed the majority of workers. Mechanization reduced employment in agriculture and mining, while clerical, professional, healthcare, education, and government jobs expanded.
Internet and Digital Networks	The wave began around the early 1990s and lasted through ~2005, with earlier seeds planted before. By 2000, about half of US households had internet access.	Invesmtents from the mid-1980s began to show in productivity beginning in early- to mid-1990s, and accelerated in the late 1990s. Labor productivity accelerated from about 1.5% per year to about 3.0% by 2000.	Digital technologies shifted the composition of business investment toward intangible assets like software, databases, algorithms, brand capital, and intellectual property.	Routine middle-skill and manufacturing occupations contracted. New occupations—software development, data science, cybersecurity, platform-based gig work, etc —expanded rapidly.

Source: Morgan Stanley Research

Six Key Takeaways for the AI Wave

If history is any guide – a big ask, we know – we point to several lessons from prior innovation waves:

1. Productivity Gains Are Likely: AI, like prior general-purpose technologies, should raise output per worker, particularly if combined with organizational redesign.

2. Labor Displacement Is Transitional, Not Permanent: Past fears of mass technological unemployment did not materialize. Job composition changes, some jobs die away, others emerge, but employment persists.

3. Boom-Bust Cycles Are Probable: Heavy AI infrastructure investment (compute, data centers, power centers) resembles prior railroad and telecom buildouts. Financial excess and volatility are likely.

4. Inequality Risks Are Elevated: AI's scalability and data network effects may intensify concentration. Without policy intervention, returns to capital and top talent could widen gaps. Income and wealth inequality are already as high as they have been in the last 125 years. We think the risk of social and political divisions remain elevated.

5. Education and Re-skilling Will Be Decisive: Every successful transition required expanding education, but also altering how education is provided. After the labor market, the sector that may end up adjusting the most is education. AI may require lifelong learning systems, not just traditional degree pathways.

6. Policy Matters: Antitrust, social insurance, and human capital investment shaped outcomes in prior waves. Institutional adaptation determines whether gains are broadly shared.

What if History Doesn't Rhyme?

The goal of this report was to find commonalities across innovation waves that could provide insight into today's environment, but we acknowledge that this time could be different. As mentioned above, diffusion timelines have been accelerating, and faster AI adoption could lead to more disruption than in the past. Others have cited that AI could have a unique breadth across industries given it is considered a general-purpose technology.

In the extreme, If AI is indeed different, then it may substitute for labor rather than augment it (Korninek 2024, Korinek and Stiglitz 2021, Trammell and Korinek 2023). In this scenario, productivity could accelerate as it automates both production and innovation, large-scale displacement of labor could occur, and education would shift toward continuos re-skilling. Rather than experience standard business cycles, the economy could experience a regime shift to higher growth while inequality sharply increases from a falling labor share.

We do not rule out that AI could defy historical precedents and create more extreme outcomes. That said, we think history is a good guide in helping us form baseline expectations of what is to come while acknowledging potential risk scenarios as well.

The First Wave: The Industrial Revolution (late 18th to mid-19th century)

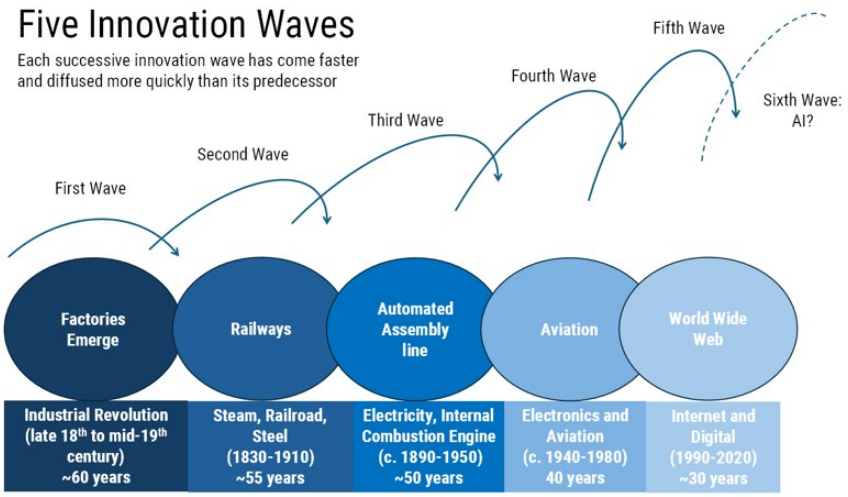
Technological progress has been a driving force behind US economic growth and transformation since the nation's founding. History reveals a sequence of major innovation waves, each centered on a cluster of breakthrough technologies, that fundamentally altered the structure of the U.S. economy and society. From the first Industrial Revolution in the early 1800s to the late-20th-century digital revolution, these waves of innovation have spurred rapid increases in productivity and output, redefined how and where people work, and raised living standards over time.

The innovation waves also brought periods of disruption: old industries declined as new ones emerged, workers' skills fell into or out of demand, and communities and policymakers struggled to adjust.

This report examines five pivotal innovation waves in U.S. economic history, with a focus on their transformative technologies, effects on business investment and infrastructure, impacts on the labor market and skills, boom-bust dynamics, changes in productivity, and broader social and political responses (including evolving patterns of inequality, public sentiment toward technological change, and shifts in education).

Understanding these historical episodes provides a valuable perspective on what looks to be the sixth major innovation wave in the US – artificial intelligence – and how its benefits and challenges might be navigated.

The Five Innovation Waves in the US over the past 250 Years



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Technological Breakthroughs and Duration

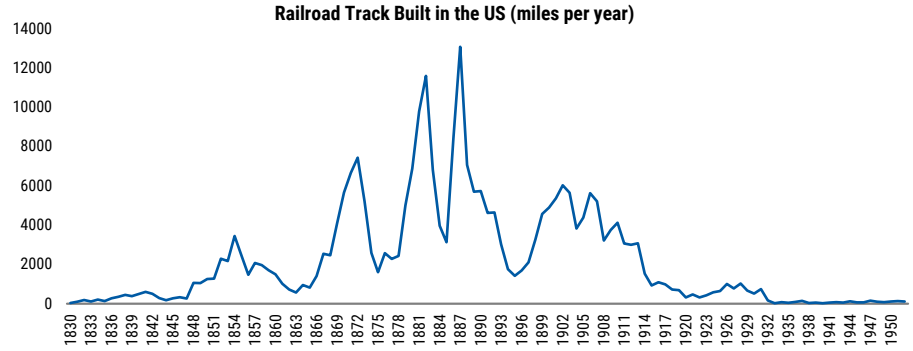
The first industrial revolution marked America's transition from a predominantly agrarian, artisan economy to one defined by mechanized industry (Ware 1931, Hunter 1979, and Dublin 1979). Its origins can be traced to the late 1700s, when water-powered textile mills were established in New England. Most economic historians trace the beginning of the

first industrial revolution to 1793, when Samuel Slater built the first successful cotton-spinning mill in Pawtucket, Rhode Island.

Over the span of about 50 years, American entrepreneurs adapted and improved British innovations to suit local conditions, laying the groundwork for industrialization. The period from approximately the 1790s through the 1840s saw the spread of steam power and factory-based manufacturing, although progress was gradual at first. Early factories relied heavily on water power and thus clustered in river-based regions of the Northeast (Atack et. al. 1980).

The first U.S. railroads also appeared during this era, heralding a new age of steam-powered transportation. By 1830, just one year after Britain's first railway, the United States had 23 miles of rail track, which surged to over 2,800 miles by 1840. Taken together, innovations in textiles, high-pressure steam engines, canals and railroads, and iron-making defined the initial industrial revolution in America.

Exhibit 2: Annual US railroad track construction began in 1830 and accelerated into the late 1890s



Source: FRED Database, NBER, Morgan Stanley Research

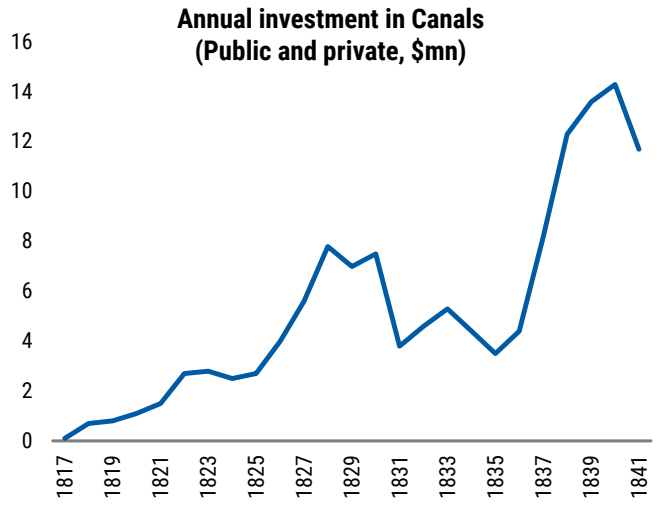
Infrastructure investment: Factories and transportation infrastructure

Although the U.S. remained a predominantly rural nation in the early 19th century, the industrial revolution began to reorient the economy from agriculture toward manufacturing and industry. The spread of factories dramatically increased the scale and efficiency of producing goods, moving production out of homes and small workshops into centralized mills. Cotton textiles became America's first major factory industry, especially in New England, benefiting from southern cotton supplies and protective tariffs against British imports. The federal government played a role in nurturing early industry by contracting with manufacturers, which supplemented private demand and kick-started production at private firms.

Equally transformative were improvements in transportation infrastructure. The early 19th century saw massive investment in turnpikes, canals, and steamboats, followed by railroads from the 1830s onward. These projects required significant capital (much of it raised by state governments or private investors) and often led to speculative booms and busts.

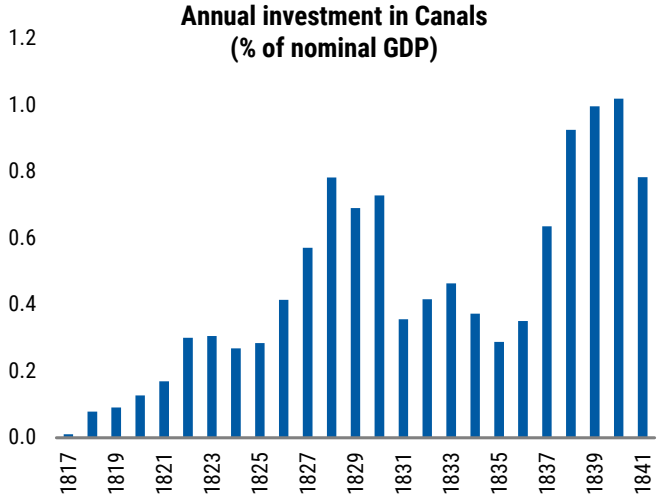
For example, canal construction boomed in the 1810s–1820s, only for many canal projects to collapse in the financial panic of 1837. Total public and private investment in canals rose steadily from 1817 through 1840, with expenditures equal to about 1.0% of GDP per year at its peak (Cramer 1960). This is equivalent to about $315bn in current dollar terms based on the BEA's advance estimate of 4Q 2025 US GDP.

Exhibit 3: Investment in canals rose steadily from 1817 to its high in 1840..



Source: Cranmer, Jerome H., 1960, Canal Investment, 1815-1860. In *Trends in the American Economy in the Nineteenth Century*, Princeton University Press, Morgan Stanley Research

Exhibit 4: ...amounting to about 1% of GDP annually at its peak



Source: Cranmer, Jerome H., 1960, Canal Investment, 1815-1860. In *Trends in the American Economy in the Nineteenth Century*, Princeton University Press, Morgan Stanley Research

Labor markets: The birth of nonfarm employment

The first industrial revolution profoundly altered the US labor market, though the process was gradual and regionally uneven. In 1800, the vast majority of Americans lived on farms or relied on craft production. By 1860, over half of the labor force still worked in agriculture, but the farm share had declined to roughly 53 percent as industrial and service employment expanded in cities (Atack, Bateman, and Weiss 1980; Carter et al. 2006). Lebergott (1966) reports similar figures. In 1800 he shows that 74% of Americans worked in agriculture. [1] By 1850, the share of farm employment slipped to 55%, while nonfarm employment stood at 45%. By the mid-1850s, the absolute number of workers engaged primarily in agriculture had begun to level off, while employment in manufacturing and construction more than doubled between 1820 and 1860 (Atack and Bateman 1999).

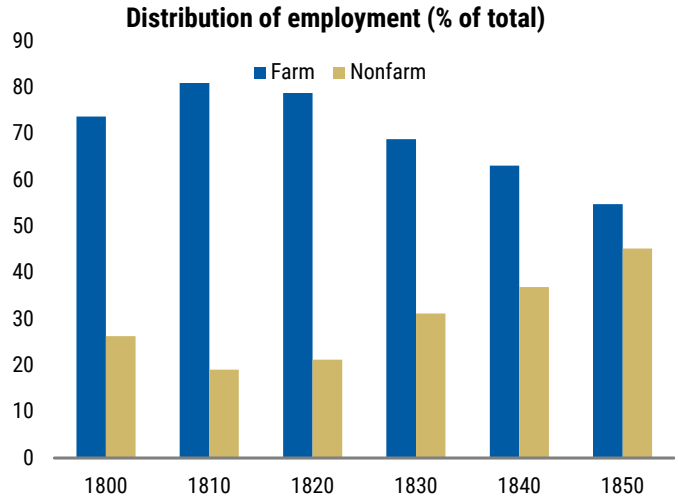
One distinct alteration in the labor market was the shift from a skilled workforce to low-skilled or semi-skilled workforce. The new factories hired unskilled or semi-skilled workers, including women and children, to operate machines, sometimes displacing traditional artisans and home-based craftsmen. For example, water-powered textile mills in New England used relatively unskilled labor, including young farm women recruited under the Lowell system, to produce cloth at scale, reducing demand for skilled

1. Some of the differences in employment shares across academic studies may be a result of definitions, including employment and the labor force. For example, Lebergott (1966) estimates of the labor force were made based on the total number of males aged ten and over. Notwithstanding definitional differences, the trend from farm to nonfarm employment is clear across all sources.

handweavers (Dublin 1979; Ware 1931). In Britain, mechanization reduced the number of handloom weavers from roughly 250,000 around 1800 to fewer than 10,000 by mid-century, and a similar process occurred in the US (Allen 2009). Crafts like shoemaking, furniture-making, and metalworking faced increasing competition from factory output.

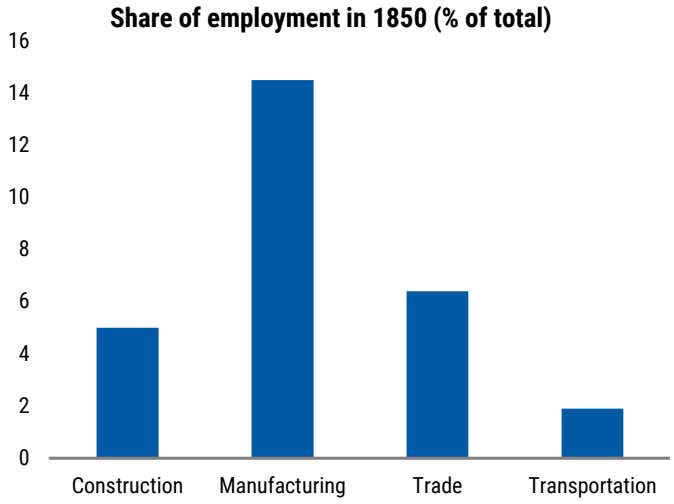
In aggregate, however, industrialization created jobs even as it destroyed others. In other words, innovation caused a churn in labor markets, but labor demand did not evaporate. Instead, employers demanded different skillsets than before. Employment opportunities grew in urban mills, tool shops, ironworks, rail yards, and coal mines, largely drawing labor from agriculture. To meet the rising demand for labor, Immigration accelerated (it played a greater role during the second innovation wave), with many seeking and finding work in industrializing cities (Carter et al. 2006). This first wave of mechanization thus initiated America's century-long transition from an agricultural to an urban-industrial labor force.

Exhibit 5: As the first industrial revolution proceeded, the share of nonfarm employment increased...



Source: Lebergott, Stanley, 1966, Labor Force and Employment, 1800-1960. In *Output, Employment, and Productivity in the United States after 1800*, Dorothy S. Brady, ed. NBER, Morgan Stanley Research

Exhibit 6: ...with new jobs coming in construction, manufacturing, trade, and transportation



Source: Lebergott, Stanley, 1966, Labor Force and Employment, 1800-1960. In *Output, Employment, and Productivity in the United States after 1800*, Dorothy S. Brady, ed. NBER, Morgan Stanley Research

Productivity: Gradual adoption equals modest productivity gains

The macroeconomic performance of the US at this time reflected gradual technological diffusion combined with westward expansion. Estimates of real GDP per capita suggest modest gains of roughly 0.5–0.7% per year between 1790 and 1830, accelerating to about 1.1% annually between 1830 and 1870 (Maddison 2001; Carter et al. 2006). That said, much of this growth stemmed from population increases, westward settlement, procurement of western lands, and capital accumulation rather than large economy-wide efficiency gains from industrialization (North 1961).

Still, key sectors and regions registered notable productivity improvements. Steam engines expanded power per worker and per machine, raising labor productivity and capital utilization in transportation and manufacturing (Hunter 1979; Atack, Bateman, and

Weiss 1980). Mid-century manufacturing shows measurable gains in labor productivity and total factor productivity linked to mechanization and the "American System" of standardized, interchangeable parts, which reduced reliance on skilled artisans and enabled larger-scale output (Hounshell 1984; Atack and Bateman 1999).

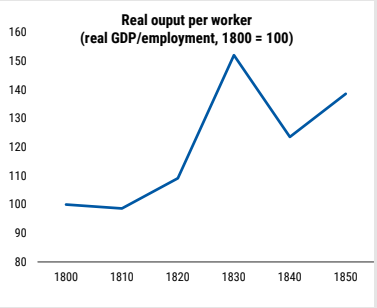
Higher productivity eventually translated into rising output, real wages, and living standards, though early gains were uneven. In both Britain and the United States, real wages rose only slowly relative to output per worker during the initial decades of the industrial revolution. In Britain this became known as the "Engels' Pause," implying a temporary decline in labor's share based on observations made by Friedrich Engels in 1845 (Allen 2009; Lindert and Williamson 1983). However, by the mid-19th century at the end of the first industrial revolution, US real wages were clearly trending upward, and manufactured goods such as textiles and tools became cheaper and more widely accessible (Lindert and Williamson 1983; Carter et al. 2006).

Box 1. A back-of-the-envelope estimate for productivity growth during the industrial revolution

Perhaps not surprisingly, official productivity data during the late 1700s and early 1800s are hard to come by. To provide some insight into what may have occurred, we construct two simple measures of output per person. In the first, we define productivity as real GDP divided by employment. In the second, we divide real GDP by the estimated labor force. Data on real GDP come from MeasuringWorth, while the labor force and employment data come from the Census Bureau (*Historical Statistics of the United States: Colonial Times to 1970*). At this time, the labor force was defined as persons 10 years old and over.

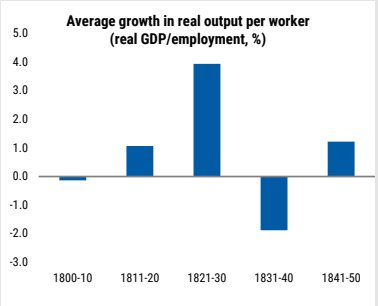
The data reveal similar estimates, with real output per person employed rising by an average of 0.84% per year between 1800 and 1850. When defined in terms of the labor force, output per worker rose by 0.96% per year. Gains across decades were uneven due, in part, to variation in population and labor force growth and the panics of 1819 and 1837 which caused substantial slumps in economic output.

Exhibit 7: Using employment data, real output per worker rose by 57% from 1800 to 1850, or just under 1.0% per year...



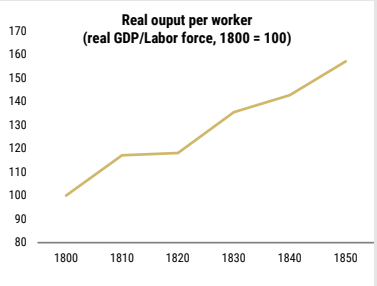
Source: Census Bureau, MeasuringWorth, Morgan Stanley Research

Exhibit 8: ...though gains were uneven across decades due, in part, to rapid labor force growth



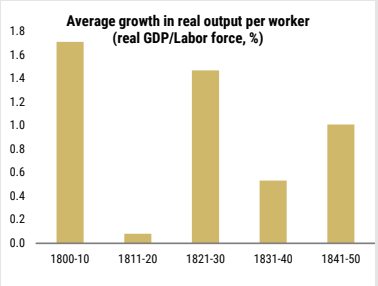
Source: Census Bureau, MeasuringWorth, Morgan Stanley Research

Exhibit 9: Using labor force data, real output per worker rose by 57% from 1800 to 1850, or about 1.0% per year...



Source: Census Bureau, MeasuringWorth, Morgan Stanley Research

Exhibit 10: ...though gains were uneven across decades due to the panics of 1819 and 1837



Source: Census Bureau, MeasuringWorth, Morgan Stanley Research

Boom-bust cycles: Geographic concentration limited contagion, but some large-scale investments did not pay off

The industrial revolution saw the first nationwide financial crises in the Panic of 1819 and 1837. The Panic of 1819 followed a post–War of 1812 credit and land boom fueled by rapid bank lending and high cotton prices, while the Panic of 1837 was deeper and more prolonged, emerging from speculative land sales, among other items (Temin 1969; Rothbard 1962; Rousseau 2002).

Importantly, in our view, neither panic was directly related to industrial mechanization in the New England factory districts. Early American industrialization remained geographically concentrated and modest in macroeconomic weight relative to agriculture and land markets before mid-century (Atack and Bateman 1999). Instead, both crises were rooted primarily in credit cycles tied to land speculation, cotton export volatility, and weaknesses in the early banking system under hard money constraints.

That said, the industrial revolution saw large-scale network investments succumb to overoptimistic demand projections and fragile financing structures, which were exposed during the panics. The canal boom of the 1820s and 1830s produced many unsuccessful ventures. Pennsylvania, Ohio, Indiana, and Illinois undertook ambitious programs financed by heavy borrowing. Pennsylvania's Main Line of Public Works and the Illinois and Michigan Canal, for example, imposed large fiscal burdens and contributed to state debt crises after 1837 (Goodrich 1960; Wallis 2000). Several states – including Arkansas, Mississippi, and Florida – defaulted on debts in the early 1840s following the collapse in land values and canal revenues (Wallis 2000).

Not all early infrastructure projects failed, however. The Erie Canal, completed in 1825, was successful, sharply reducing transport costs and generating toll revenues that exceeded construction costs (North 1961; Goodrich 1960).

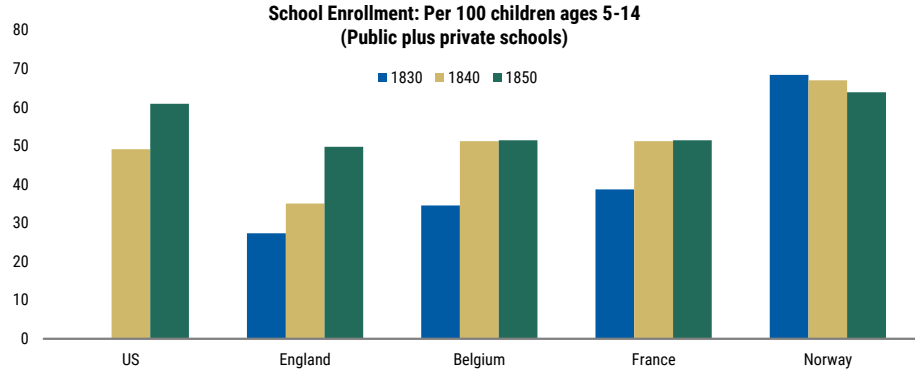
Social and Political Responses: Income inequality and worker rights

Industrialization generated predictable distributional conflict. In Britain, the Luddite episodes of the 1810s reflected skilled artisans' resistance to deskilling and capital

deepening (Hobsbawm 1952; Allen 2009). In the United States, craft workers in the 1820s and 1830s organized early trade unions and citywide labor associations, pressing for shorter hours and higher wages and encountering adverse judicial rulings (Wilentz 1984). These responses illustrate a recurring pattern: technological change raises aggregate productivity while redistributing income across skills and sectors.

At the same time, demographic and geographic adjustment accelerated. The US population nearly quadrupled between 1800 and 1850, and the urban share rose from roughly 6 percent to 20 percent by 1860 (Carter et al. 2006). Urbanization generated economies of scale that still exist today, but also congestion costs, prompting municipal public health and sanitation reforms. Contemporary observers such as Tocqueville noted widening disparities between industrial wealth and wage labor, highlighting concerns about inequality, child labor, and concentrated economic power (Tocqueville 1835). These distributional tensions foreshadowed the more organized labor and agrarian political movements of the later 19th century.

Exhibit 11: By the end of the industrial revolution, educational enrollment was in line with European peers



Note: Data for Norway is public school enrollment only. Source: 1840 and 1850 US censuses, Lindert (2004, vol. 2, App.Table A.1), Fishlow (1966a, Table 1)

Education and Skills Development: Increased returns to human capital

The industrial revolution led to the expansion of the common school movement, reflecting the view that industrialization increased returns to basic human capital. Horace Mann and other reformers argued that tax-supported primary education would promote social mobility and supply a more disciplined workforce. By the 1840s, northern states had established statewide public elementary systems with standardized curricula and teacher training institutions (Goldin 1998). Although early factory production required limited formal schooling, the industrialization of the Northeast placed growing emphasis on literacy, numeracy, and machine operation.

Public school enrollment expanded through the 1850s, while institutions such as Rensselaer Polytechnic Institute, founded in 1824, signaled nascent demand for applied scientific and engineering training. Still, educational access remained uneven, particularly in the South, and most workers had little beyond primary schooling. Substantial expansion

of secondary and higher education would follow only in later innovation waves, when technological change more decisively raised the skill premium.

The Second Wave: Steam, Railroads, and Steel (1830-1910)

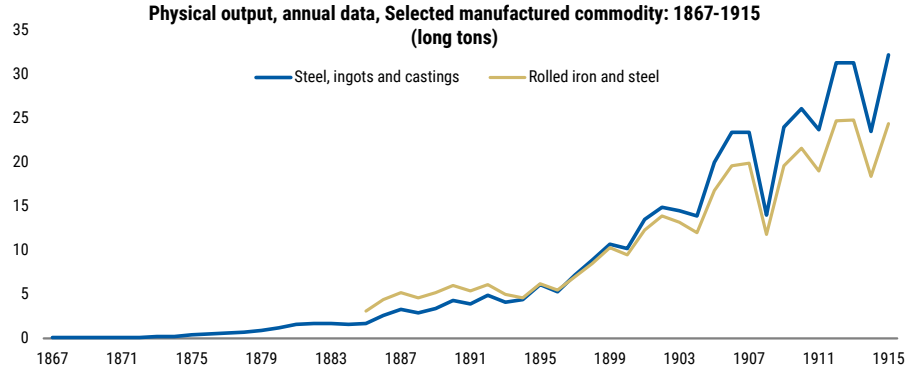
Technological Breakthroughs and Timeline

The second innovation wave in US history unfolded from the 1830s through the early 20th century. Often termed the "Second Industrial Revolution," this innovation wave was driven by steam power, railroads, steel production, and heavy machinery. The late 19th century also witnessed major advances in communications. Many enabling technologies of the modern industrial economy either emerged or reached maturity in this period, including:

- The **railroad network** expanded spectacularly after the US Civil War. Track mileage grew from about 30,000 miles in 1860 to roughly 192,000 miles by 1900, connecting regions into an integrated national market (Fogel 1964). The completion of the **first transcontinental railroad** in 1869 sharply reduced coast-to-coast travel times and accelerated western settlement.
- **Steam power** became widespread in both transportation and manufacturing, freeing factories from dependence on water power and enabling urban industrial growth (Chandler 1977).
- The development of **mass-produced steel** revolutionized construction and manufacturing. Steel rails lowered railroad costs, while steel was increasingly used in bridges, machinery, ships, and skyscrapers.
- The **commercial telegraph, the telephone, and early electrification** reduced coordination costs and transformed business organization. The commercial telegraph (1840s), the telephone (patented by Alexander Graham Bell in 1876), and Thomas Edison's inventions (from the phonograph in 1877 to the incandescent light bulb in 1879) all belong to this era.

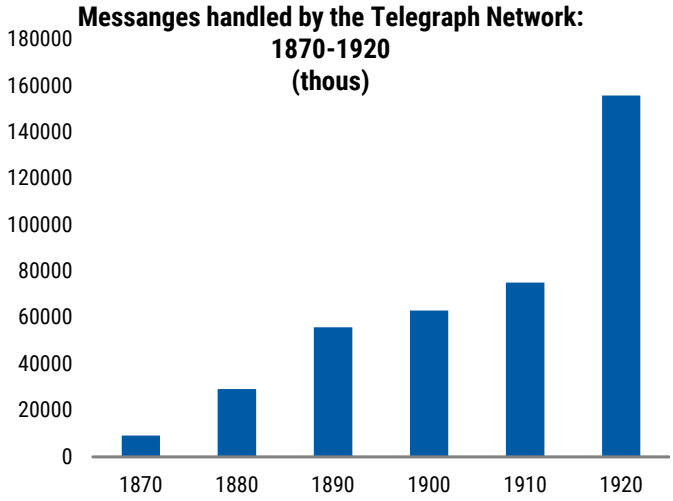
The steam and steel revolution propelled the United States into the front ranks of industrial powers. Between the Civil War and World War I, US output expanded dramatically. By the early 20th century the United States had surpassed Britain as the world's leading manufacturer. Railroad construction absorbed vast quantities of capital, financed through stock and bond issuance and often supported by federal land grants. By the early 1900s, innovations in transportation, communications, and heavy industry were diffusing nationwide, fundamentally altering business operations and daily life.

Exhibit 12: US steel production rose steadily during the second innovation wave



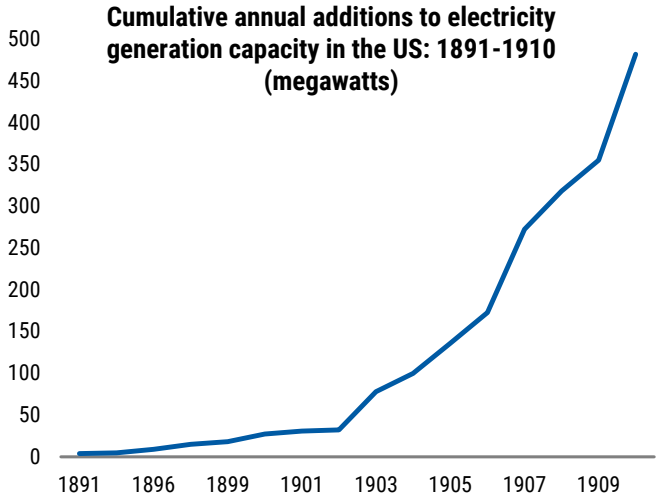
Note: A long ton is also known as an imperial ton British ton, or gross ton and is a unit of mass equal to 2240 pounds or 1016 kilograms.
Source: Census Bureau, Morgan Stanley Research

Exhibit 13: Telegraph messages rose from 9.2mn in 1870 to 155.7mn by 1920



Source: Census Bureau, Morgan Stanley Research

Exhibit 14: Electricity generation capacity accelerated rapidly in the early 1900s



Note: Hydro provided all cumulative annual additions to electricity generation capacity in the US until 1915, when oil gained prominence. Source: US Energy Information Administration, Morgan Stanley Research

Labor Market Changes and Urbanization: Manufacturing nets one quarter of the workforce

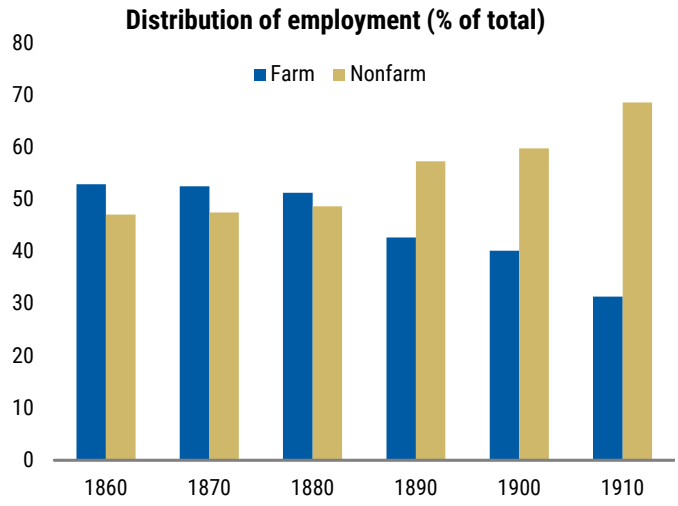
Between 1850 and 1910, the labor market experienced extraordinary reallocation. Agriculture's employment share fell steadily—from over half of the workforce in 1860 to roughly 30 percent by 1910—while manufacturing and related industries expanded (Carter et al. 2006). By the early 20th century, manufacturing employed roughly one-quarter of the workforce. Steel, meatpacking, machinery, garments, and food processing absorbed millions of workers, many of them recent immigrants. More than 18 million immigrants arrived between 1880 and 1914, supplying labor to factories and railroads.

New white-collar occupations also emerged. Clerks, telegraph and telephone operators, bookkeepers, and managers became increasingly important as firms grew in scale and complexity. The rise of the modern corporation marked what Chandler (1977) termed the

managerial revolution. Urbanization accelerated: by 1900, approximately 40 percent of Americans lived in cities, up from 20 percent in 1860 (Carter et al. 2006). Industrial hubs such as Chicago, Pittsburgh, and Cleveland expanded rapidly.

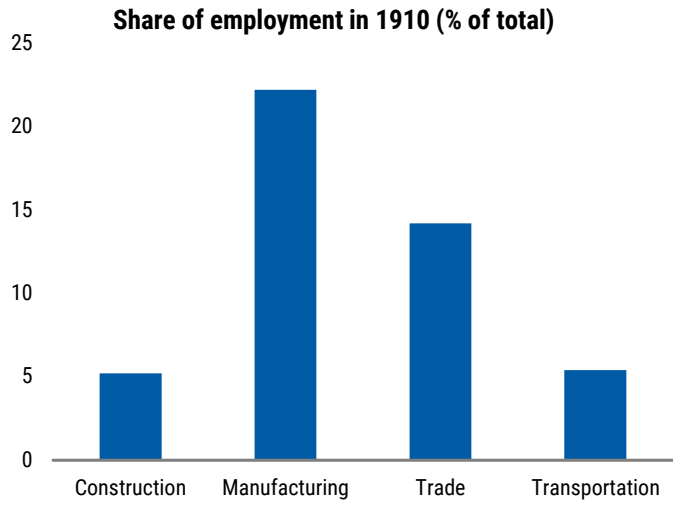
At the same time, certain occupations declined. The expansion of railroads and urban transit reduced demand for traditional transport trades tied to horse-drawn vehicles. Although unemployment was often elevated during financial panics and recessions, long-run labor force growth continued. Technological change repeatedly displaced specific jobs but generated new ones elsewhere, producing net employment expansion over extended horizons.

Exhibit 15: Nonfarm employment exceeded agriculture employment by 1890...



Source: Lebergott, Stanley, 1966, Labor Force and Employment, 1800-1960. In *Output, Employment, and Productivity in the United States after 1800*, Dorothy S. Brady, ed. NBER, Morgan Stanley Research

Exhibit 16: ...with manufacturing and trade accounting for 36% of employment



Source: Lebergott, Stanley, 1966, Labor Force and Employment, 1800-1960. In *Output, Employment, and Productivity in the United States after 1800*, Dorothy S. Brady, ed. NBER, Morgan Stanley Research

Wages, Living Standards, and Inequality: Enter the Gilded Age

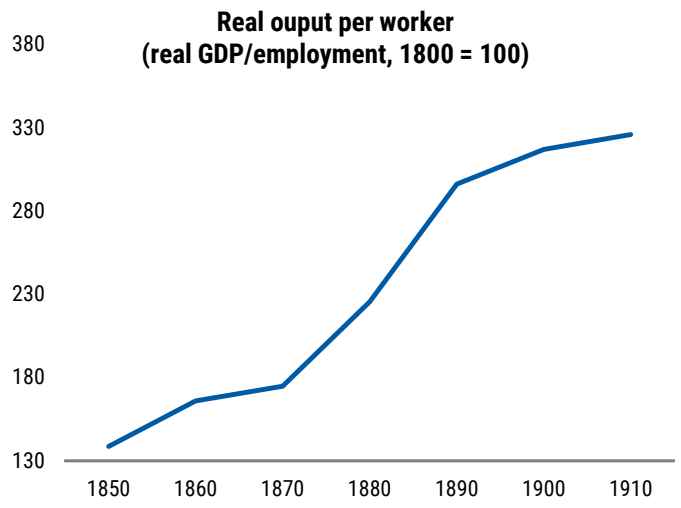
The age of steam and steel produced substantial gains in productivity and income. By the late 19th century, labor productivity growth approached 2% annually, or about double the rate of the first industrial revolution, leading to a sustained rise in real GDP per capita (Maddison 2001). In the Exhibits below, we find similar results. A simple estimate of real output per person – using either employment data or labor force data – suggests productivity gains of 2.1-2.3% per year in the second half of the 1800s. Railroads broadened access to goods, urban transit improved mobility, and communications technologies compressed economic distance.

Yet these gains were unevenly distributed due, in part, to industry concentration and monopoly power. The Gilded Age witnessed extreme wealth concentration, with leading industrialists amassing vast fortunes while many workers faced precarious employment and poor working conditions (Piketty and Saez 2003). Financial panics in 1873, 1884, 1893, and 1907 intensified social tensions. Major labor conflicts, including the Homestead Strike

(1892) and Pullman Strike (1894), reflected distributional conflict between capital and labor.

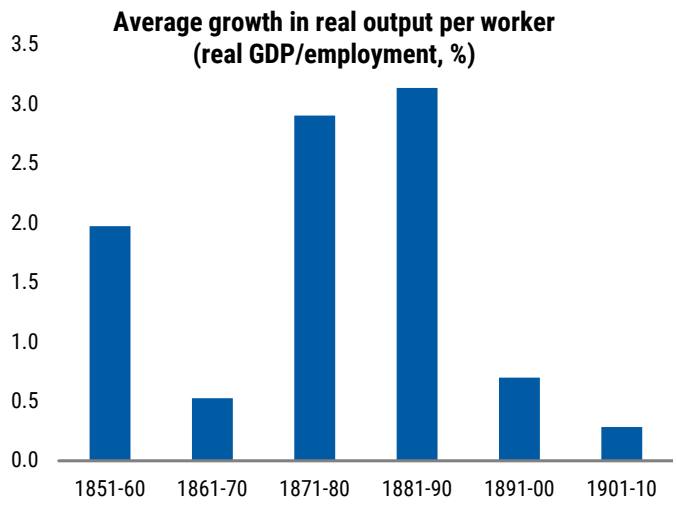
These pressures contributed to Progressive Era reforms. Antitrust legislation, railroad regulation, early labor protections, and the introduction of a progressive federal income tax after 1913 marked a significant expansion of federal oversight of industrial capitalism (Kolko 1965; Hovenkamp 1991).

Exhibit 17: Using employment data, real output per worker more than doubled between 1850 and 1910



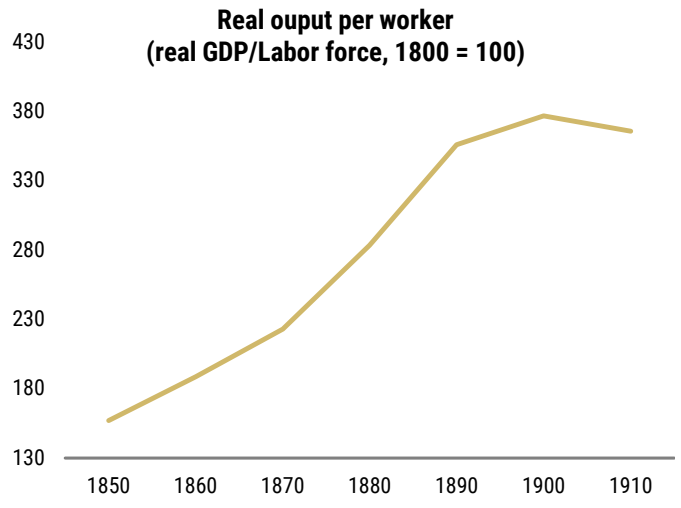
Source: Census Bureau, MeasuringWealth, Morgan Stanley Research

Exhibit 18: The second half of the 1800s saw average annual productivity gains of 2.1%



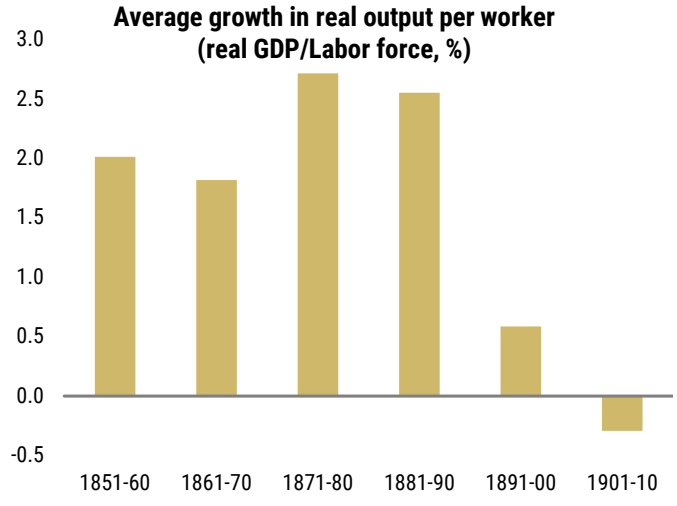
Source: Census Bureau, MeasuringWealth, Morgan Stanley Research

Exhibit 19: Using labor force reveals a similar picture: real output per worker more than doubled between 1850 and 1910



Source: Census Bureau, MeasuringWealth, Morgan Stanley Research

Exhibit 20: Using labor force reveals a similar picture: the second half of the 1800s saw average annual productivity gains of 2.3%



Source: Census Bureau, MeasuringWealth, Morgan Stanley Research

Boom-bust: Capital-intensive heavy industry

This rapid economic expansion contributed to financial excess and periodic crashes, including the Panics of 1873 and 1893, both closely linked to railroad overbuilding and fragile financing (Kindleberger and Aliber 2005; Romer 1989). These crises were not incidental to industrialization, but they were tightly connected to the financial architecture of the second innovation wave.

Railroads were the largest capital-intensive enterprises of the era, requiring unprecedented fixed investment in track, rolling stock, and terminals. From 1872-1881, railroad investment averaged 2.5% of GDP – about $790bn in current dollars – and railroad investment was routinely 10% or more of gross capital formation (Cootner 1963). If there is any historical antecedent to the burgeoning AI-spending story, it is railroad-related investment.

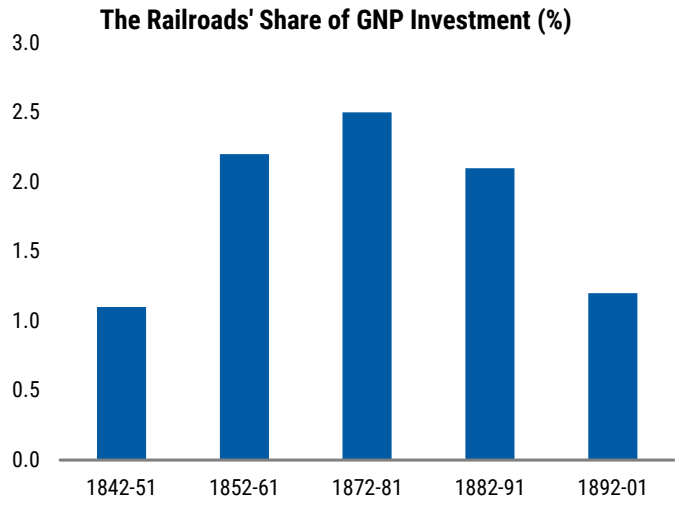
Financing relied heavily on bond issuance marketed domestically and in Europe, often backed by optimistic projections of westward settlement and freight demand (Fogel 1964; Chandler 1977). When traffic revenues failed to meet expectations or foreign capital flows reversed, highly leveraged railroad balance sheets proved vulnerable.

The Panic of 1873 followed the failure of Jay Cooke & Co., a major underwriter of Northern Pacific bonds, triggering widespread railroad bankruptcies and a prolonged contraction. The Panic of 1893 similarly reflected overexpansion and rate competition that eroded profitability, leaving many lines insolvent. In both episodes, industrial concentration increased through reorganization, as financiers consolidated distressed assets into larger systems, reinforcing the rise of corporate capitalism (Lamoreaux 1985; Chandler 1977).

More broadly, the scale and indivisibility of steam, steel, and rail technologies amplified cyclical volatility. Large fixed costs, network externalities, and expectations of national market integration encouraged rapid expansion during booms and sharp retrenchment during downturns. We interpret the financial panics of the late 19th century as endogenous byproducts of industrialization's capital intensity and speculative financing, rather than exogenous shocks unrelated to technological change.

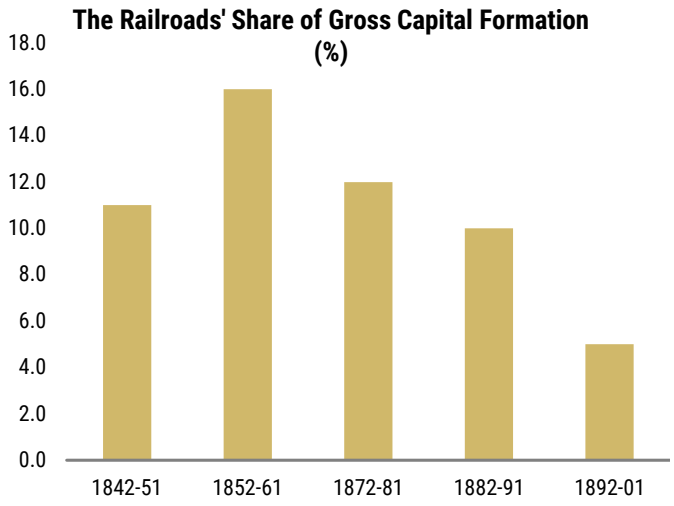
Following the panics, heavy industry concentrated. Carnegie's steel operations achieved large economies of scale, while merger waves around 1900 created U.S. Steel under J.P. Morgan. In oil, Standard Oil came to dominate refining through horizontal integration and control of transportation networks (Chandler 1977).

Exhibit 21: For five decades, average railroad investment as a share of GDP exceeded 1%, with a peak at 2.5%



Source: Cootner, Paul H., 1963, "The Role of Railroads in United States Economic Growth," The Journal of Economic History, Vol. 23, No. 4, December, 477-521. Morgan Stanley Research

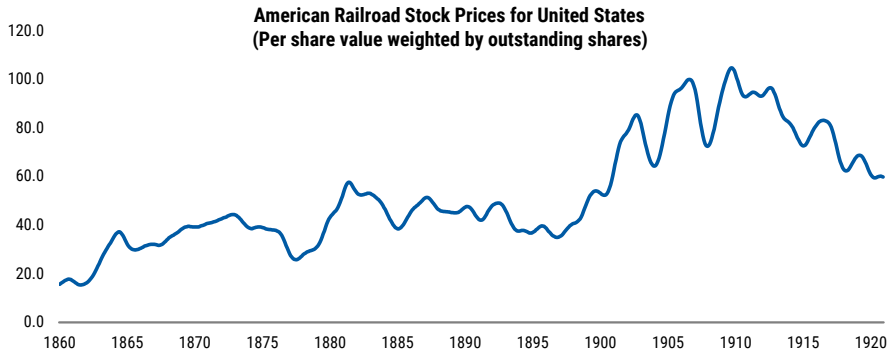
Exhibit 22: Railroad investment frequently exceeded 10% of gross capital formation



Source: Cootner, Paul H., 1963, "The Role of Railroads in United States Economic Growth," The Journal of Economic History, Vol. 23, No. 4, December, 477-521. Morgan Stanley Research

Exhibit 23:

Railroad stock prices exhibited periods of boom-bust behavior



Source: NBER, American Railroad Stock Prices for United States, FRED database, Morgan Stanley Research

Changes in Skills and Education: The Land-Grand Acts

The age of Steam and Steel increased demand for engineers, technicians, and managers capable of operating large-scale industrial systems. The Morrill Land-Grant Acts of 1862 and 1890 established institutions of higher learning that emphasized engineering, agriculture, and mechanical arts, broadening access to technical education. Land-grant universities such as Iowa State (1862), Kansas State (1863), Cornell University (designated New York's land-grant institution in 1865), and Texas A&M (1876) institutionalized applied scientific training, while the 1890 Act supported historically Black colleges and universities including Tuskegee Institute and North Carolina A&T (Thelin 2011; Geiger 2015). These institutions broadened access to technical education and helped professionalize engineering and agricultural science.

By the early 20th century, the high school movement expanded secondary education

enrollment dramatically, equipping workers with literacy, numeracy, and clerical skills required in an urban-industrial economy (Goldin 1998; Goldin and Katz 2008). These educational expansions complemented technological change and facilitated adaptation to a more complex economic structure, building upon earlier education reform during the industrial revolution.

The Third Wave: Electricity and the Internal Combustion Engine (c. 1890-1950)

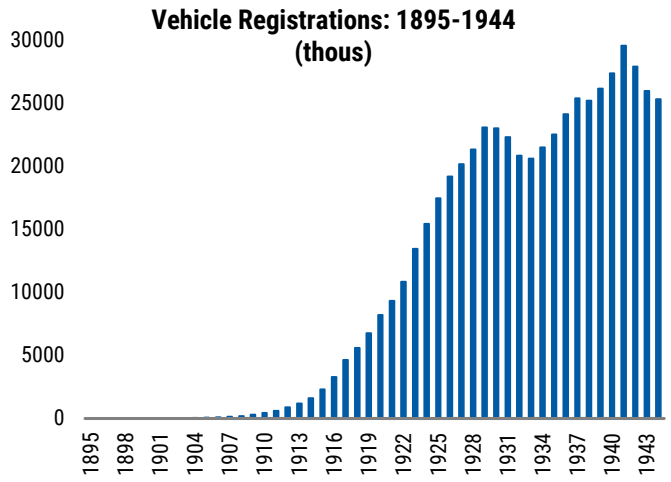
Diffusion and Organizational Change

The third innovation wave, which economic historians generally date from the 1890s through the 1940s, centered on electricity, the internal combustion engine, chemicals, and modern communications. A key feature of the third innovation wave is that the economy-wide effects of some of its main inventions, like Thomas Edison's first commercial power plant (Pearl Street Station) and early gasoline automobiles in the 1890s, manifested only with large-scale diffusion and organizational redesign (David 1990; Gordon 2016). Electrification spread rapidly after 1900 as central generating stations and transmission networks expanded from major cities to smaller towns. Electricity generation increased more than tenfold between 1902 and 1929, reflecting both industrial and household adoption (Gordon 2016). The shift from steam-driven line shafts to decentralized electric motors allowed for more flexible plant layouts and continuous-flow production, raising productivity and facilitating mass production.

The internal combustion engine similarly transformed transportation and spatial organization. Ford's Model T (first produced in 1908) and its moving assembly line for mass-producing automobiles (introduced at Ford's Highland Park plant in Michigan in 1913) dramatically reduced automobile prices and assembly time, catalyzing car ownership during the 1920s. By 1930, the US had 23mn registered vehicles, which necessitated large-scale road construction, oil pipeline expansion, and suburban development.

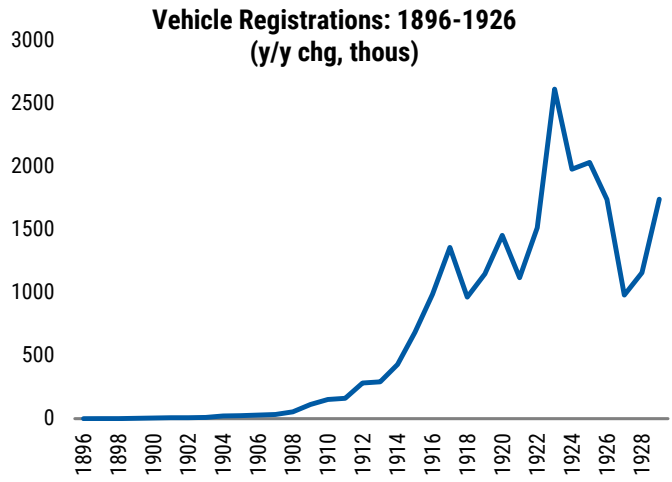
Other innovations during this time period included electrified urban transit; household appliances like refrigerators, radios, and vacuum cleaners; and telecommunications further reshaped daily life and industry. These technologies were broad, foundational innovations that affected nearly every part of the economy and worked best when combined with changes in many different industries.

Exhibit 24: Vehicle registrations in the US surged to 23mn by 1930 and 30mn by 1944



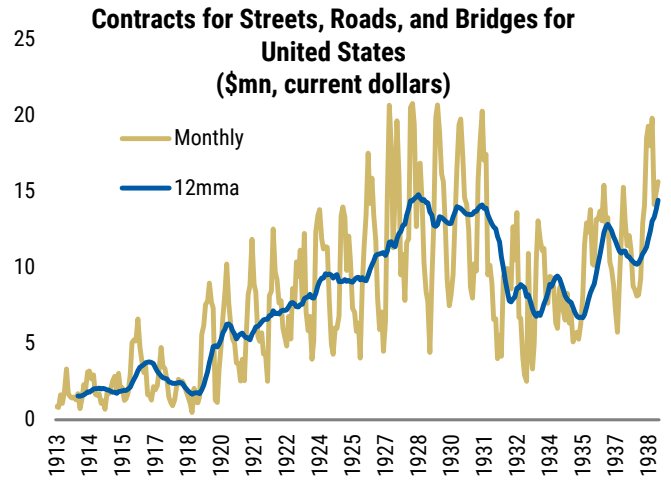
Note: Passenger cars, total for the United States. Source: NBER, FRED database, Morgan Stanley Research

Exhibit 25: The US registered an additional 1-2mn vehicles per year from 1916-29



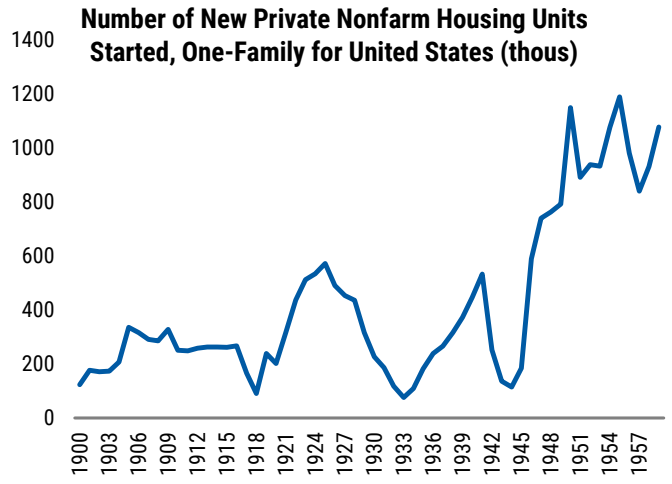
Note: Passenger cars, total for the United States. Source: NBER, FRED database, Morgan Stanley Research

Exhibit 26: New cars needed new roads, and contracted spending moved steadily higher...



Source: NBER, Fred database, Morgan Stanley Research

Exhibit 27: ...with the automobile also facilitating suburban expansion, especially after 1945



Source: NBER, Fred database, Morgan Stanley Research

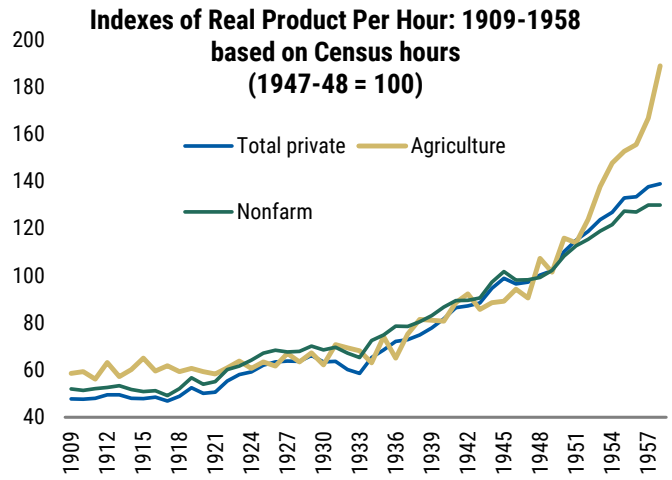
Productivity and Growth: The "Great Leap"

Business investment in electric utilities, automotive plants, petroleum refining, and chemical production surged during the third innovation wave. Utilities built coal-fired power stations and transmission grids nationwide, while oil displaced coal as the dominant fuel source for transport. Electrification and motorization contributed to what has been termed the "Great Leap" in productivity: growth in total factor productivity averaged about 1.5 percent annually from 1909 to 1929. Between 1889 and 1953, the real net national product of the US grew from less than $20bn to $187bn (in 1929 prices). This nearly tenfold increase over sixty four years represents an average annual compound rate of growth of more than 3.5% (Kendrick 1961). Real output per hour in nonfarm business

approximately doubled between 1900 and 1929. Despite the severe contraction of the 1930s, by the end of World War II productivity levels far exceeded those of the pre-1900 economy.

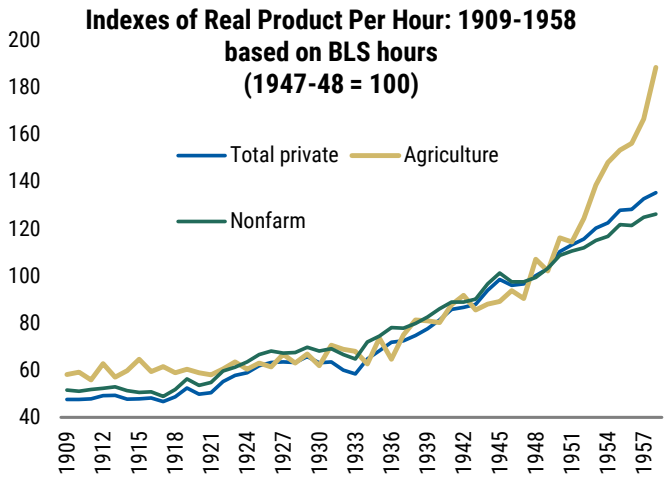
The data support roughly three cycles in the movement of output per hour between 1909 and 1958. Troughs in productivity occurred around 1917, 1933, and 1945, which correspond to the end of WW1, the Great Depression, and WW2. Still, despite these major geopolitical and macroeconomic events, productivity averaged a solid 2.0% growth rate in the private sector, with a ten-year moving average ranging between 1.0% and 3.5%.

Exhibit 28: Output per hour based on Census estimates of hours worked



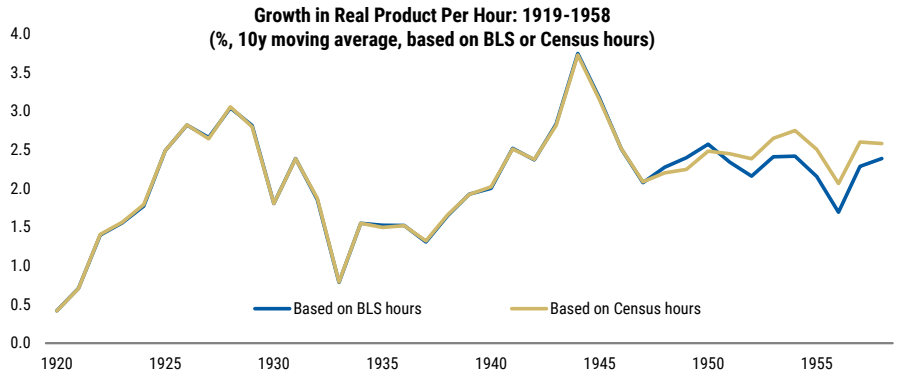
Source: BLS, FRB St. Louis, Morgan Stanley Research

Exhibit 29: Output per hour based on BLS estimates of hours worked



Source: BLS, FRB St. Louis, Morgan Stanley Research

Exhibit 30:
Growth in real output per hour for the nonfarm private sector averaged 32.0% between 1909 and 1958



Source: BLS, FRB St. Louis, Morgan Stanley Research

Booms and busts: Pronounced cyclical volatility

This era, however, was marked by pronounced cyclical volatility. Indeed, the third innovation wave includes the Great Depression, the defining macroeconomic event of the 1900s. The 1920s saw speculative booms in utilities, automobiles, consumer durables, and

radio, fueled by easy credit, rising stock market valuations, and widespread optimism about a permanently higher growth path. Investment flowed into electric holding companies, automobile manufacturers, and related suppliers, often financed through leverage and complex financial structures.

The 1929 stock market crash abruptly reversed these expectations and was followed by the Great Depression. While economists generally cite financial fragility, the banking sector collapse, and monetary contraction as causes of the Depression rather than the result of an innovation cycle, the capital intensity and rapid expansion of electrified and motorized industries likely amplified the booms and busts (Temin 1976). High fixed costs, heavy borrowing, and interconnected supply chains meant that when demand fell, firms cut production sharply, deepening the contraction and transmitting distress across sectors.

Labor Markets: A shift toward professional occupations

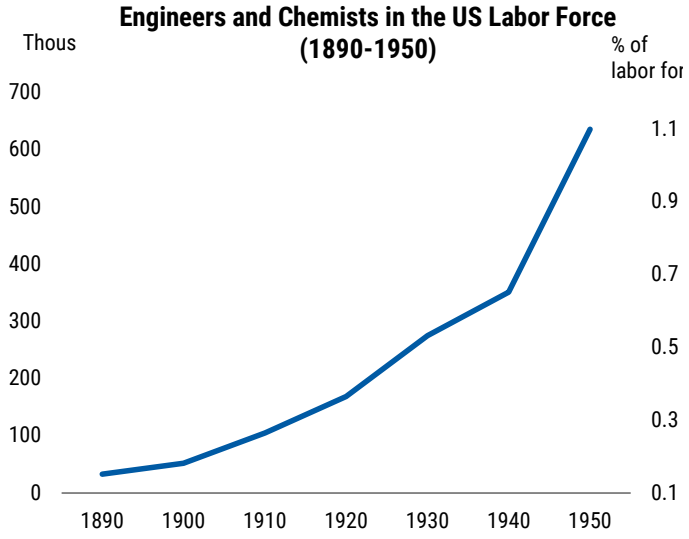
Electrification and mechanization accelerated structural transformation that was already underway in the US. As we wrote about here, agriculture's employment share fell from roughly one-third in 1900 to about one-fifth by 1940, as tractors, combines, and trucks reduced labor demand on farms (Carter et al. 2006). Manufacturing employment expanded in absolute terms through mid-century, while clerical and administrative occupations grew rapidly with the rise of large corporations. Between 1910 and 1950, the clerical workforce roughly tripled, and by the mid-20th century white-collar workers outnumbered blue-collar workers. Assembly-line production increased demand for semi-skilled operatives even as certain craft occupations declined.

Just as it is today, innovation-led unemployment became a recurrent concern, particularly during the Great Depression when unemployment approached 25 percent. Yet over longer horizons, job creation in new sectors offset losses in older ones, and working hours declined substantially. Unionization expanded under the Wagner Act (1935), contributing to rising wages in mass-production industries. [2]

The pronounced cyclical volatility of the era was also evident in the evolution of the unemployment rate. Even if we exclude the Great Depression, the period from 1900-1950 saw 10 different recessions based on dating tradition from the NBER. Excluding the Great Depression, the unemployment rate ranged from a low of 1.0% in 1906 to a high of 11.9% in 1921. Importantly, in our view, the natural rate of unemployment was relatively unchanged over the period. Using simple smoothing techniques from the Hodrick Prescott filter, we find that the natural rate of unemployment was 3.1% in 1901 and 3.5% by 1950. Although the economy went through historically large swings, the third innovation wave did not alter the basic matching characteristics of the labor market: those unemployed for cyclical reasons were able to find re-employment over time.

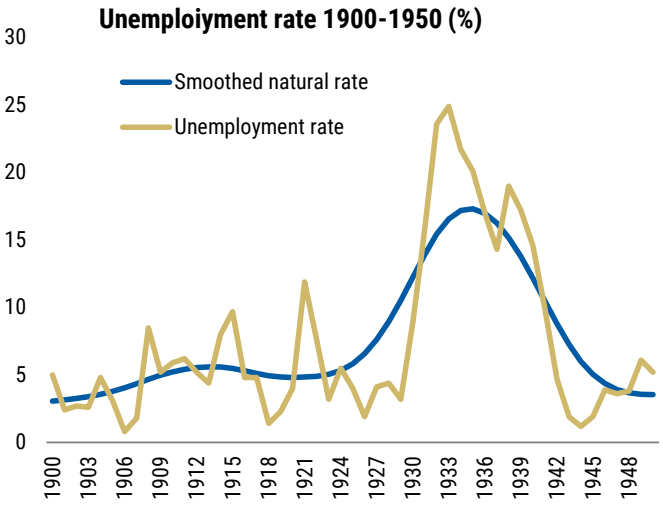
2. The Wagner Act (1935), formally known as the National Labor Relations Act (NLRA), was a cornerstone New Deal law that fundamentally reshaped labor relations in the United States. It guaranteed workers the right to unionize and engage in collective bargaining, prohibited unfair labor practices by employers, such as firing workers for organizing or interfering with union activity, and created the National Labor Relations Board (NLRB) to enforce labor law and oversee union elections. The Act shifted the legal balance toward labor, leading to a rapid expansion of union membership, particularly in mass-production industries like autos, steel, and electrical manufacturing. Unionization rose sharply after 1935, contributing to higher wages, more stable employment contracts, and the mid-20th-century "Great Compression" in income inequality (Goldin and Margo 1992).

Exhibit 31: Over the sixty year period from 1890-1950, the total number of engineers and chemists in the workforce increased eighteen-fold



Source: Kendrick, John W., 1961, Productivity and Economic Growth, in *Productivity Trends in the United States*, Princeton University Press and NBER. Morgan Stanley Research

Exhibit 32: The third innovation wave saw heightened cyclical volatility in the economy and labor market, though without an upward trend in NAIRU



Source: Lebergott, Stanley, 1957, "Annual Estimates of Unemployment in the United States, 1900-1954," In The Measurement and Behavior of Unemployment, Universities – National Bureau Committee for Economic Research. Morgan Stanley Research

Exhibit 33:
The shift toward higher-paying industries in the early 1900s was associated with relative shifts in the labor force toward more highly skilled or professional occupations, and from farms to nonfarm pursuits

Social-Economic Distribution of the Civilian Labor Force, 1890-1950				
Group	1890	1910	1930	1950
Nonfarm	37.4	69.0	79.0	88.1
Proprietors, managers, etc.		6.6	7.5	8.8
Professional persons	3.7	4.4	6.1	8.6
Clerks and kindred workers	6.0	10.2	16.3	19.3
Skilled workers and foremen		11.7	12.9	14.2
Semiskilled workers		14.7	16.4	20.3
Unskilled workers		21.4	19.8	16.9
Farm	42.6	31.0	21.0	11.9
Proprietors, managers	23.1	16.5	12.4	7.5
Laborers	19.5	14.5	8.6	4.4

Source: Kendrick, John W., 1961, Productivity and Economic Growth, in *Productivity Trends in the United States*, Princeton University Press and NBER. Morgan Stanley Research

Politics, Inequality, and Education

The social response to the third innovation wave combined enthusiasm for higher living standards with political reform. The New Deal reshaped labor and social policy through Social Security, collective bargaining protections, and public works programs that expanded infrastructure, including rural electrification (Temin 1976). Income inequality narrowed during the "Great Compression" from the 1940s through the 1970s, supported by progressive taxation, strong unions, and expanding education (Goldin and Margo 1992). The third innovation wave is unusual in this regard; most innovation waves in the US

resulted in greater income inequality, not less.

Human capital formation accelerated. The high school movement dramatically increased secondary enrollment between 1910 and 1940, far outpacing Europe (Goldin 1998; Goldin and Katz 2008). Land-grant and public universities that began in the second innovation wave expanded engineering and applied science training, while the GI Bill (1944) greatly broadened access to higher education and homeownership. By mid-century, mass education complemented electrification and motorization, ensuring that productivity gains translated into rising middle-class incomes and sustained economic growth.

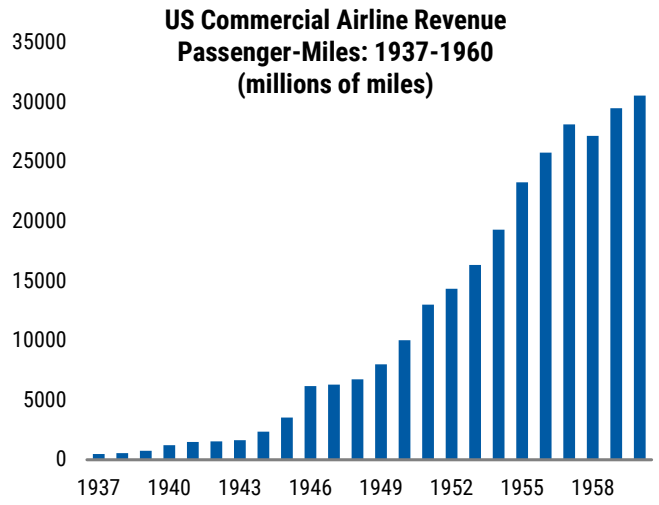
The Fourth Wave: Electronics and Aviation (c. 1940-1980)

Technological Foundations: A successful public-private partnership

The fourth innovation wave, which economic historians generally date from the 1940s through the 1980s, was propelled by advances in electronics, aerospace, and early digital computing. Not surprisingly, many of the innovations that altered the private sector of the US economy during this time period were rooted in World War II mobilization. Federal government spending through research and development outlays during the war and subsequent Cold War accelerated radar, jet propulsion, nuclear engineering, and computing.

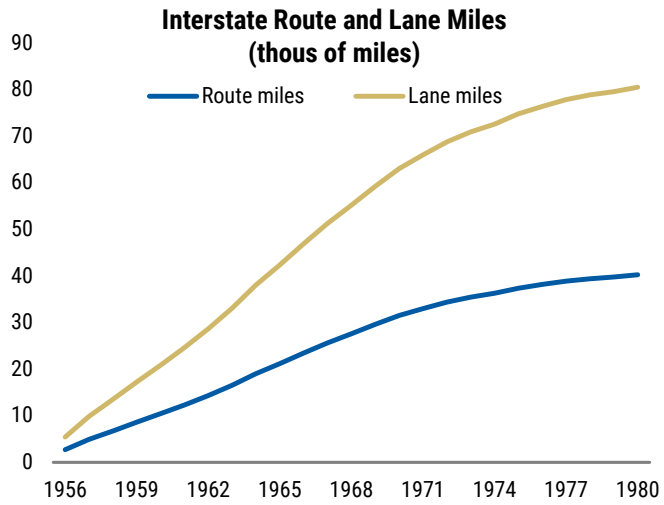
An important inflection point came in 1947 with the invention of the transistor and subsequent development of integrated circuits. Both laid the foundation for modern electronics. At the same time, commercial jet aviation (for example, the Boeing 707 was released in 1958) dramatically reduced travel times and further increased global integration (Gordon 2016). The Interstate Highway Act of 1956 further reorganized economic geography, facilitated suburbanization, and established long-haul trucking and the seeds of just-in-time logistics.

Exhibit 34: Commercial airline miles flown increased following WW2



Note: Revenue passenger miles for US commercial carriers only. Source: FAA Statistical Handbook of Aviation. Morgan Stanley Research

Exhibit 35: Interstate mileage expanded rapidly between 1956 and 1980

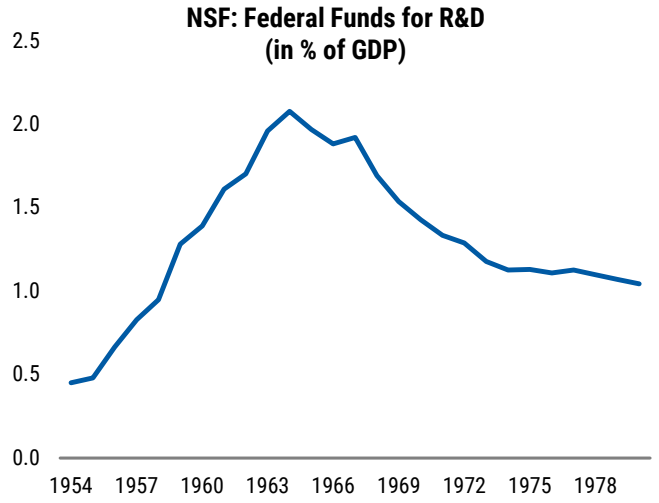


Note: Lane miles equals route miles x number of lanes. Source: FHWA Highway Statistics, Morgan Stanley Research

These technologies were complementary to earlier electrification and motorization. Television diffused rapidly in the 1950s and 1960s, reshaping advertising and politics; mainframe computers entered business and government data processing; and defense-related projects seeded spillovers into civilian sectors. The era was exemplified by a public-private partnership between the federal government and the private sector that led to a four-decade period of innovation. The result was an increasingly science-based

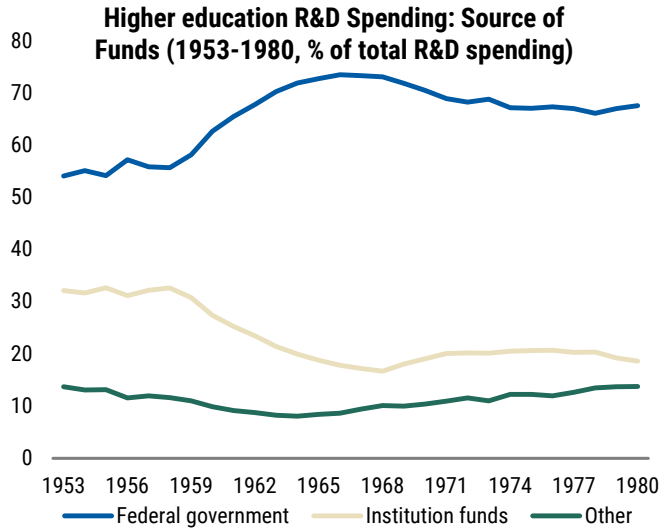
production system, with corporate research laboratories and federally funded university research playing central roles (Chandler 1977).

Exhibit 36: Federal R&D Spending increased four-fold as a share of GDP from 1954 to 1964



Source: BEA, NCSES, NSF, Morgan Stanley Research

Exhibit 37: The Federal Government was the largest source of R&D spending in Higher Education



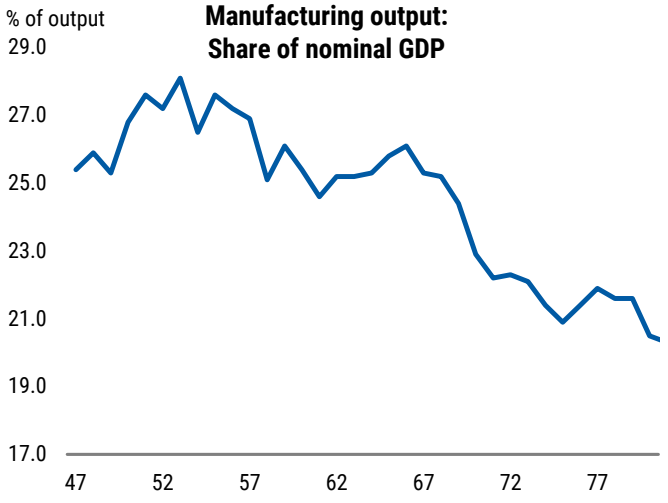
Note: Other includes S&L government, Business, and other. Source: NCSES, NSF, Morgan Stanley Research

Labor Markets: A full-on shift toward services employment

The postwar decades coincided with a structural shift toward services. This may sound at odds with the performance of manufacturing during the early post-war period, where manufacturing output continued to rise and the US dominated many important manufacturing industries. However, in terms of employment, manufacturing's share of employment peaked in the late 1940s and began a gradual decline that, in many ways, continues to this day.

By 1970, service-providing sectors employed the majority of American workers. That said, unionized blue-collar workers in autos, steel, and durable goods enjoyed rising real wages during the "blue-collar middle class" era, even as demand grew for engineers, programmers, technicians, and aerospace specialists. Mechanization further reduced employment in agriculture and mining, while clerical, professional, healthcare, education, and government jobs expanded. These shifts marked an early stage of the transition from an industrial-based economy toward a knowledge-intensive economy.

Exhibit 38: Manufacturing output, both as a share of US GDP...

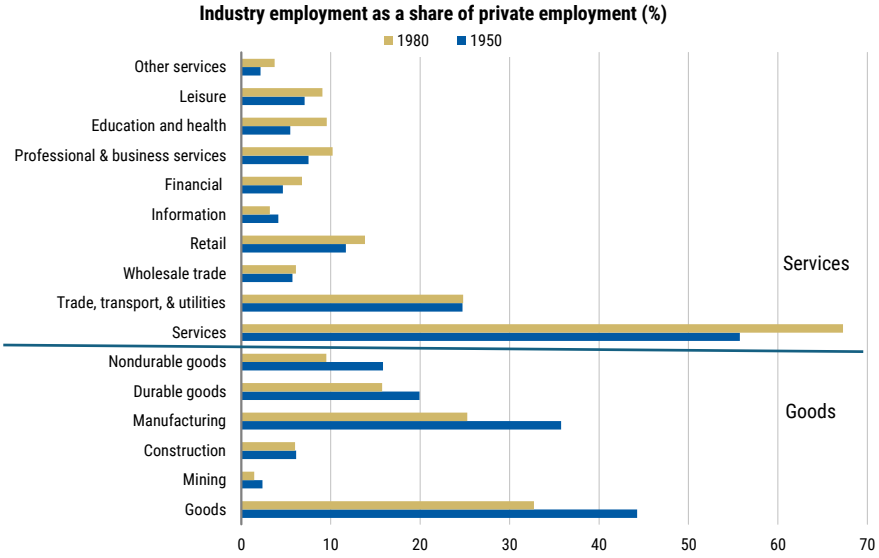


Source: BEA, Haver Analytics, Morgan Stanley Research

Exhibit 39: ...and share of employment, has steadily declined from 1950 to 1980

Source: BEA, Haver Analytics, Morgan Stanley Research

Exhibit 40:
Private sector employment shifted in the direction of services from 1950 to 1980



Source: BLS, Haver Analytics, Morgan Stanley Research

Productivity and Slowdown

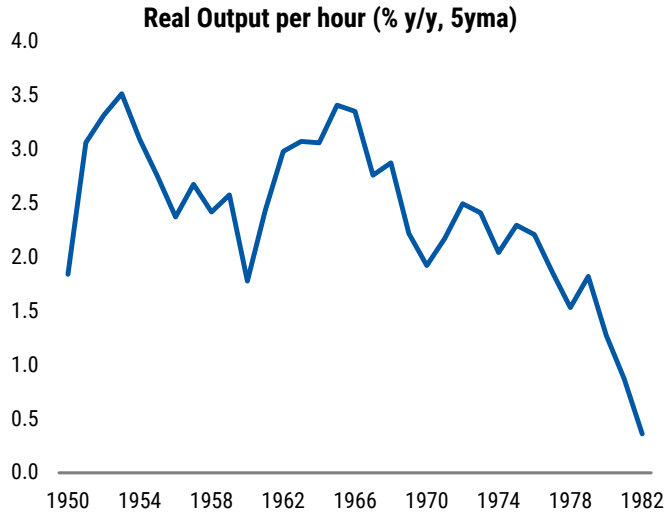
The period from 1945 to 1973 is often characterized as a "Golden Age" of growth. Real GDP growth averaged above 4 percent annually, and labor productivity grew roughly 2.5–3 percent per year, among the strongest sustained rates in US history (Gordon 2016). Capital deepening, diffusion of earlier general purpose technologies, and defense-supported innovation all contributed.

After 1973, however, productivity growth slowed markedly, which economists later

dubbed the "productivity slowdown." Gordon (2016) argues that several forces contributed to the slowdown. First, the oil shocks of 1973–74 and 1979 sharply raised energy prices (a large relative price shock to a critical input), reducing measured productivity by forcing costly adjustments in energy-intensive industries and reallocating capital toward less productive uses.

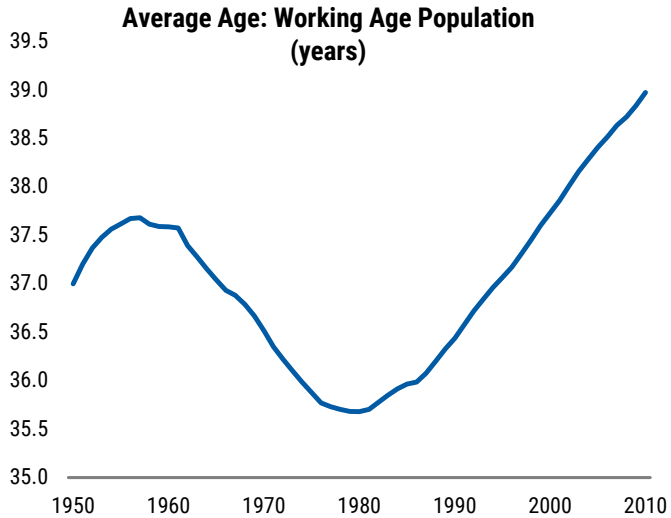
Second, Gordon saw the extraordinary one-time gains from electrification, motorization, and urban infrastructure as largely exhausted by the early 1970s. He argued that later innovations, while important, did not deliver comparable broad-based efficiency gains. Third, rising regulatory burdens in transportation, environmental policy, and workplace safety may have redirected resources toward compliance rather than output.

Exhibit 41: Productivity growth was elevated between 1950 and 1970, and then slowed sharply on account of...



Source: BEA, Haver Analytics, Morgan Stanley Research

Exhibit 42: ..full diffusion, increased regulatory burdens, relative price shocks, and a declining average working age



Source: Census Bureau, Haver Analytics, Morgan Stanley Research

Fourth, demographic shifts—including slower labor force growth and the entry of less-experienced workers—temporarily dampened average productivity. Finally, Gordon emphasizes measurement issues and transitional adjustment costs, as firms reorganized around new technologies whose benefits were initially diffuse and unevenly captured in official statistics (Gordon 2016).

Altogether, we argue the behavior of productivity during this period still supports the case for knowledge-based innovations boosting output per worker. At some point, the impulse to productivity fades as diffusion across the economy is completed and the conclusion of that impulse happened to coincide with several important external shocks that weighed on the US economy for the better part of a decade.

Inequality and Skill Premium: A tipping point toward income inequality

Income inequality remained compressed through the 1950s and 1960s, extending the trend that began during the third innovation wave and New Deal era reforms. However, the fourth innovation wave saw the end of the "Great Compression", which was

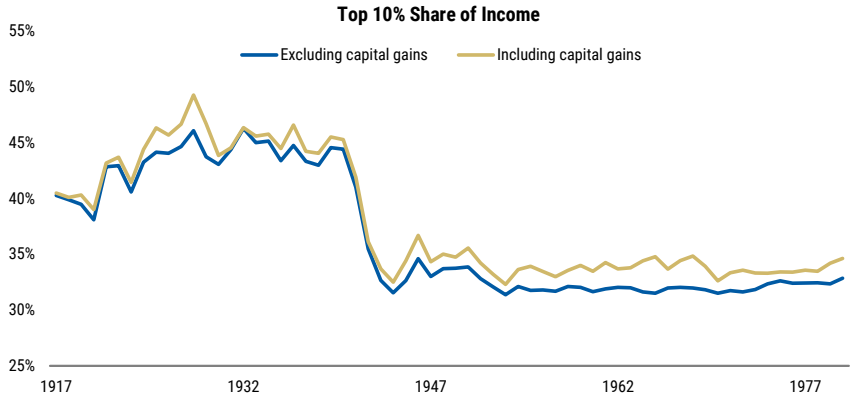
supported, in part, by strong unions, progressive taxation, high marginal tax rates, and the broad diffusion of secondary and higher education (Goldin and Margo 1992).

The institutional framework built during the third innovation wave—collective bargaining under the Wagner Act, Social Security, and mass public education—ensured that productivity gains from electrification and mass production translated into rising middle-class wages.

However, by the late 1970s this equilibrium began to shift. Wage inequality began to widen sharply in the 1980s, marking what many economic historians view as a tipping point in the postwar distributional regime. Skill-biased technological change associated with computing and early automation increased the premium to college education, while globalization and import competition placed downward pressure on routine manufacturing wages (Autor, Katz, and Krueger 1998; Autor, Dorn, and Hanson 2013).

At the same time, declining union density, deregulation, and reduced progressivity in taxation weakened the mechanisms that had previously compressed wage dispersion. The result was a sustained rise in top income shares and capital income concentration after 1980, reversing much of the mid-century equalization (Piketty and Saez 2003).

Exhibit 43: The 1950s to 1980s saw income compression. After 1980 this would shift in the direction of greater income and wealth inequality



Source: Saez, Emmanuel, 2017. "Striking it Richer: The Evolution of Top Incomes in the United States, Updated with 2017 final estimates," UC Berkeley. Morgan Stanley Research

Politics and Deregulation: A turn toward market liberalization

Public policy both shaped and responded to these shifts. Cold War defense spending sustained aerospace and electronics industries, while the GI Bill broadened access to higher education and homeownership. By the late 1970s, concerns about inflation and stagnation prompted airline and trucking deregulation and a broader turn toward market liberalization. At the same time, foreign competition from Europe and Japan intensified pressures on legacy manufacturing sectors, contributing to regional labor market dislocation in the 1970s and 1980s.

Education: A golden age for higher education

Higher education enrollment expanded sharply in the postwar decades, particularly in

engineering and scientific fields. Universities became hubs of federally funded research, linking academic science to industrial innovation. The GI Bill and subsequent federal education initiatives reinforced a mass higher-education system that supplied the technical and managerial talent required by an electronics- and aerospace-driven economy (Goldin and Katz 2008). In our view, college enrollment was also boosted by demographics, with the baby boom generation (those born between 1945 and 1965) turned college age in the 1960s.

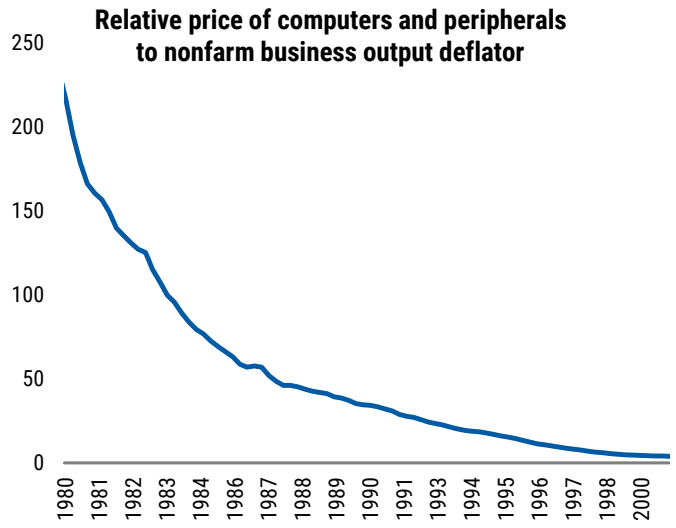
The Fifth Wave: The Internet and Digital Networks (c. 1990-2020)

Core Technologies and Duration

The fifth innovation wave, beginning in the early 1990s, is organized around digital networks, computing power, and data. The commercialization of the internet, broadband diffusion, mobile computing, and cloud infrastructure dramatically lowered the cost of storing, processing, and transmitting information. Advances in semiconductors and software expanded the scope of automation beyond routine manual tasks to include cognitive and analytical functions (Brynjolfsson and McAfee 2014; Autor 2015).

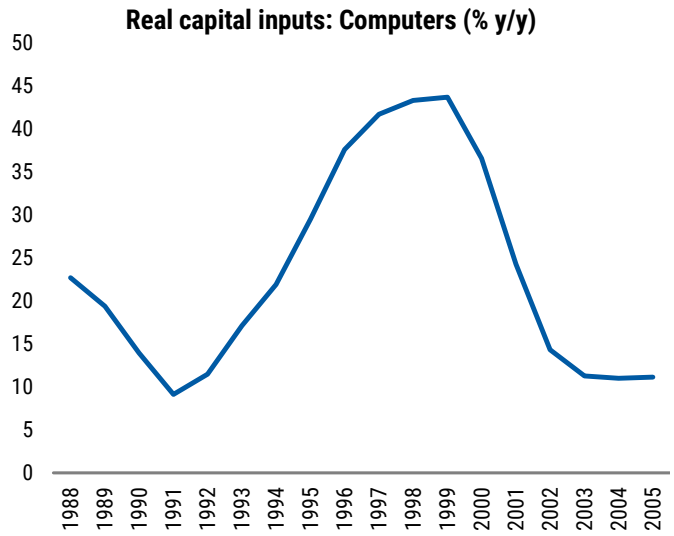
The steady decline in computing costs—often summarized by Moore's Law—enabled exponential growth in data storage and processing capacity, fostering new applications in finance, logistics, healthcare, and media (Tevlin and Whelan 2003). As in earlier waves, the largest gains required complementary organizational change—platform architectures, supply-chain digitization, data analytics integration, and new business models built around user networks and algorithmic matching. Firms that successfully reorganized around digital technologies realized outsized productivity and market-share gains, reinforcing divergence across industries and companies.

Exhibit 44: During the internet boom, the relative price of computers to business output fell rapidly...



Source: BEA, BLS, Haver Analytics, Morgan Stanley Research

Exhibit 45: ...which led to strong growth in business spending on computers and peripherals



Source: BEA, BLS, Haver Analytics, Morgan Stanley Research

The seeds of the fifth innovation wave were planted much earlier. Inventions like the transistor (invented in 1947), the integrated circuit (1958), and early mainframe computers (1940s–1950s) ultimately had their greatest transformative effects on the US economy from the 1970s through the 1990s. This era saw the diffusion of personal computers (and the founding of eventual giants Microsoft and Apple), the growth of the software industry, and the creation of the modern Internet.

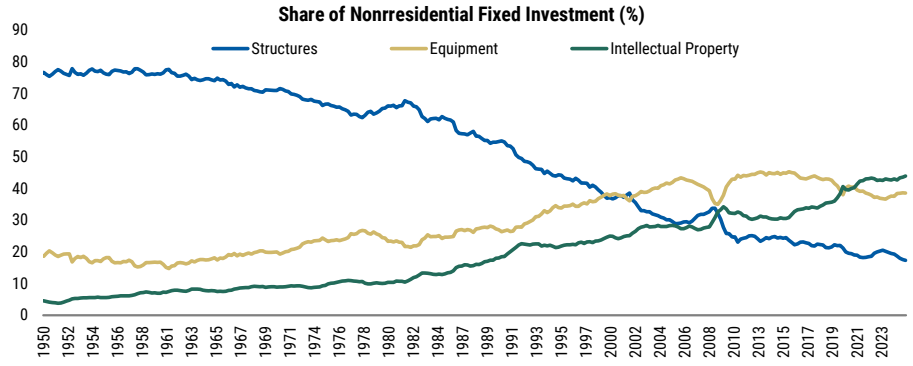
The internet was incubated in the 1960s military-academic network (ARPANET) before morphing into a commercial and mass-market network by the 1990s with the advent of the World Wide Web. By 2000, about half of US households had internet access. Other digital technologies like mobile phones and enterprise software systems further shaped this wave. In short, by the turn of the 21st century the US had entered the Information Age, where creation, processing, and transmission of information became central economic activities. The transition from an agrarian to industrial to knowledge-based economy was largely complete.

Investment shifts to Intangible Capital

During the fifth innovation wave, digital technologies further shifted the composition of business investment toward intangible assets like software, databases, algorithms, brand capital, and intellectual property. These items now represent a substantial and growing share of business investment (Corrado, Hulten, and Sichel 2005).

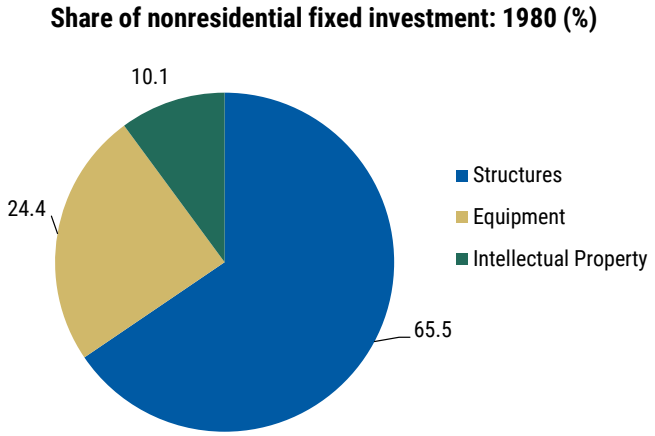
Unlike physical capital, intangible assets are scalable and often non-rival, allowing firms to serve global markets with relatively low marginal cost. Autor and others detail how platform-based firms exploit network effects, scale economies, and data feedback loops, often generating winner-take-most outcomes and high market concentration (Autor et al. 2020). Data accumulation itself has become a strategic asset, improving product quality and reinforcing competitive advantage. These features distinguish the digital era from earlier capital-intensive waves. Strategic advantage is less about physical infrastructure, which is seen primarily through the rapid emergence of US-based, globally dominant technology firms.

Exhibit 46: The share of structures in nonresidential fixed investment moved lower after 1980, replaced by larger shares of equipment and intellectual property spending



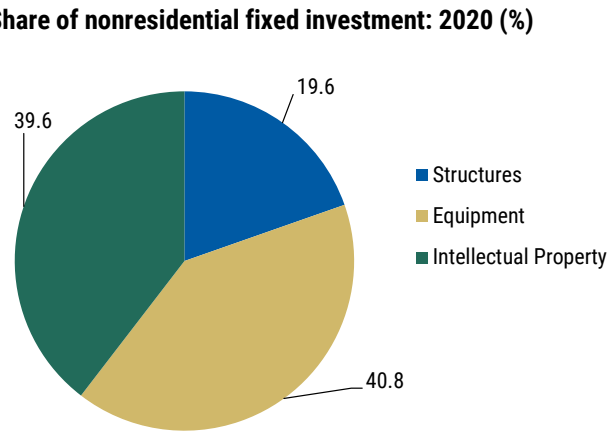
Source: BEA, Haver Analytics, Morgan Stanley Research

Exhibit 47: Structures accounted for almost two-thirds of nonresidential fixed investment in 1980



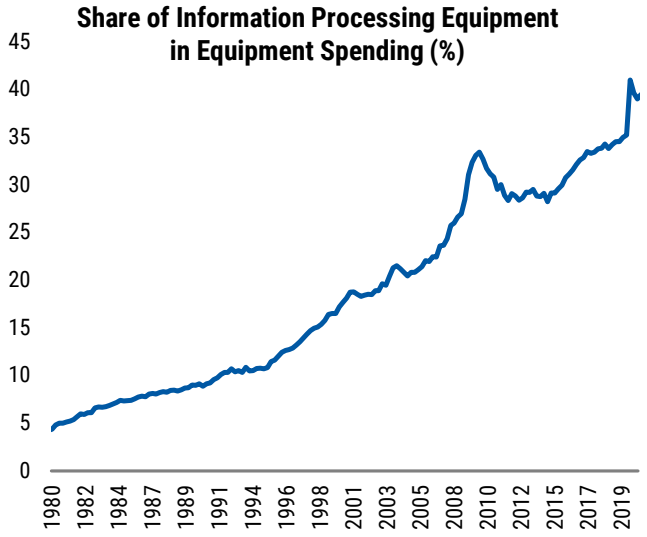
Source: BEA, Haver Analytics, Morgan Stanley Research

Exhibit 48: By 2020, equipment and intellectual property investment dominated business spending



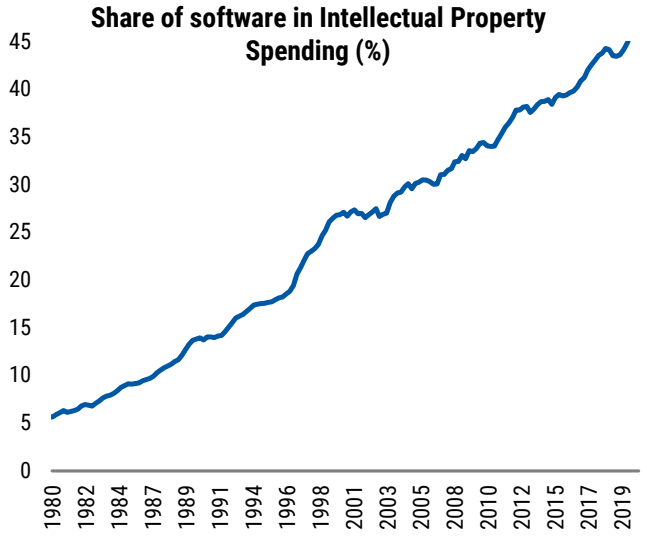
Source: BEA, Haver Analytics, Morgan Stanley Research

Exhibit 49: Spending on information processing equipment and software...



Source: BEA, Haver Analytics, Morgan Stanley Research

Exhibit 50: ...have taken up larger shares of business investment over time



Source: BEA, Haver Analytics, Morgan Stanley Research

Labor Markets: A loss of middle-skill occupations

As in the four prior innovation waves, the internet and digital revolution pushed labor markets to adjust again. Occupational data suggest routine middle-skill occupations, both blue- and white-collar, have contracted as tasks susceptible to codification and automation were either automated or offshored (Autor 2015).

These trends led many, including academic economists, to conclude that the labor market was becoming "hollowed out", with employment growth shifting toward high-skill professional and technical roles and lower-wage service jobs. The middle saw a drop in

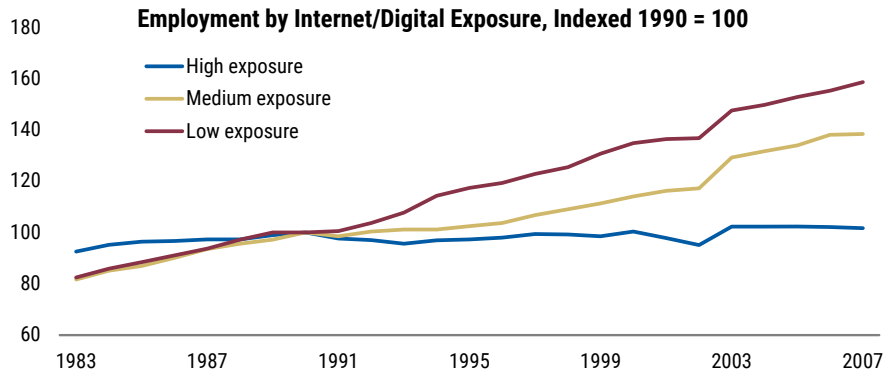
labor demand.

Manufacturing employment continued to decline sharply after 2000, reflecting a combination of automation and import competition (Autor, Dorn, and Hanson 2013). Others point to the ascension of China into the WTO and hyperglobalization as fueling these trends.

At the same time, new occupations—software development, data science, cybersecurity, digital marketing, logistics management, and platform-based gig work—expanded rapidly, illustrating the ongoing reallocation of tasks rather than a net disappearance of work, as we wrote about here. Geographic disparities widened as technology hubs attracted high-skill workers and capital, while regions reliant on routine manufacturing faced persistent adjustment costs.

In Exhibit 51 below we plot the change in employment according to internet-digital exposure from 1983 to 2007. We ranked occupations by their "routineness" and put occupations into three categories: high exposure to internet-digital automation, medium exposure, and low exposure and indexed employment in each group such that 1990 = 100. Growth in employment in low- and medium-exposure occupations outpaced that of high-exposure occupations. Occupational scores have been used by authors like Autor and Dorn (2009) to show simultaneous growth of employment in both the highest- and lowest-skilled jobs in the US labor market in recent decades.

Exhibit 51: Occupations with high "routineness" saw reduced job demand during the tech boom



Source: Autor and Dorn, 2009. Morgan Stanley Research

Productivity: A ten year burst in output per hour

Productivity growth accelerated in the late 1990s with the diffusion of information and communications technology, particularly in retail, wholesale trade, and financial services, before slowing in the 2000s. Large-scale investments in computers and peripherals beginning in the mid-1980s began to bear fruit on productivity estimates beginning in early- to mid-1990s, a subject we take up more closely in a subsequent section of this report. Labor productivity, in terms of output per hour, accelerated from about 1.5% per year to about 3.0% by 2000, before slowing back to pre-innovation wave rates by around 2005.

Explanations behind the slowdown in productivity range from measurement challenges in capturing digital output to lagged organizational adjustment and diminishing returns to

early adoption (Byrne, Fernald, and Reinsdorf 2016). Some scholars argue that intangible investments and free digital services are imperfectly reflected in GDP statistics, potentially understating true gains (the "productivity paradox").

Other economists emphasize diffusion lags and argue frontier firms experienced strong productivity growth, while lagging firms struggled to fully integrate digital tools. As in prior innovation waves, aggregate productivity performance reflects both technological potential and the speed with which complementary skills, management practices, and regulatory frameworks adjust.

Productivity accelerated, offsetting a demographic-drive slowdown in hours worked

As we introduced in our prior research (see Immigration and the Macroeconomy), we estimate potential growth using a trend-cycle decomposition framework. The framework assumes each measure of economic activity and labor markets is represented as the sum of cyclical and trend components, where the cyclical component is assumed to be common across all the inputs with contemporaneous effects. Each variable is permitted to have its own unique trend. The framework yields estimates of potential output, the building blocks of potential output, the output gap, long-run unemployment, and the equilibrium real policy rate, which are key inputs to monetary and fiscal policy.

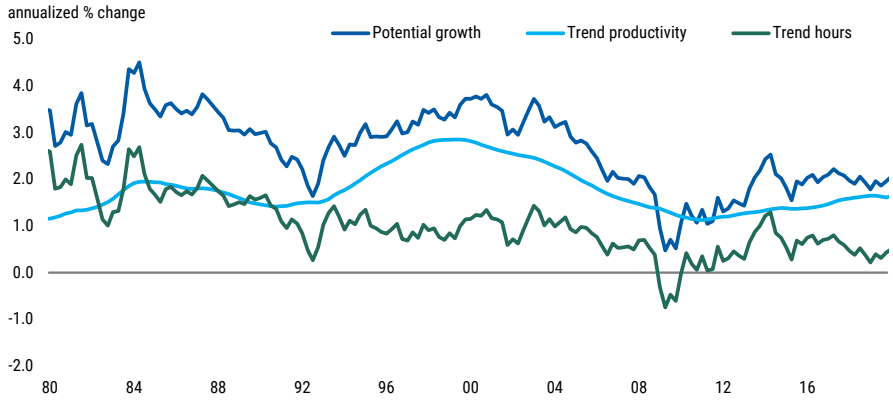
In Exhibit 52 and Exhibit 53 we show our estimates of potential growth and its inputs from 1970-2019. Beginning with output per hour, gains in trend productivity were sluggish in the 1970-1995 period, reflecting the productivity slowdown in the second half of the fourth innovation wave. However, productivity picked up thereafter in the fifth innovation wave. Our estimates clearly show the surge in productivity during the "tech boom" with trend productivity growth accelerating from about 1.5pp per year in the early 1990s to about 3.0pp by 2000, before slowing back toward 1.5pp on the eve of the global financial crisis. The productivity boom was particularly strong in the second half of the 1990s.

Exhibit 52: The tech boom of the 1990s saw a surge in productivity growth

Decomposition of Potential Output: 1970-2019						
annualized % change	1970-79	1980-89	1990-94	1995-99	2000-10	2010-19
Potential output	3.6	3.3	2.5	3.2	2.6	1.8
Total hours	2.6	1.8	1.1	0.9	0.7	0.5
Population	2.0	1.2	1.0	1.2	1.2	0.9
LFPR	0.8	0.5	0.2	0.0	-0.3	-0.4
Employment rate	0.1	0.3	0.0	-0.1	-0.1	0.1
Workweek	-0.3	-0.2	-0.2	-0.2	-0.1	-0.1
NFB labor productivity	1.3	1.6	1.6	2.6	2.0	1.4
Sector ratios						
GDO to NFBO	-0.5	-0.3	-0.3	-0.4	-0.3	-0.2
NFB to total employment	0.2	0.2	0.2	0.2	0.1	0.1

Source: Morgan Stanley Research

Exhibit 53: US GDP and its main component parts: 1980-2019



Source: Morgan Stanley Research

However, the rise in potential growth was limited by demographics. Labor-force participation (and growth in total hours) increased rapidly in the 1970s but was stagnant in the 1990s due to the deceleration in the working-age population. In the 1970s, the population contributed about 2pp to trend growth as the Baby Boomer generation entered their prime working age years. Beginning in the 1980s, the contribution from population growth slowed to around 1pp as both the rate of population growth and participation rates moderated.

The trend work week slipped throughout the entire five decade sample period, which we interpret as reflecting the ongoing shift from manufacturing to services, where the latter tends to have a shorter average work week and greater prevalence of part-time work. That said, the falling contribution of hours worked to potential – from an average of 1.8pp per year in the 1980s to 0.7pp in the 2000s – was more than offset by the rise in productivity, leading to faster potential growth.

Boom-Bust: Winners often emerge alongside losers

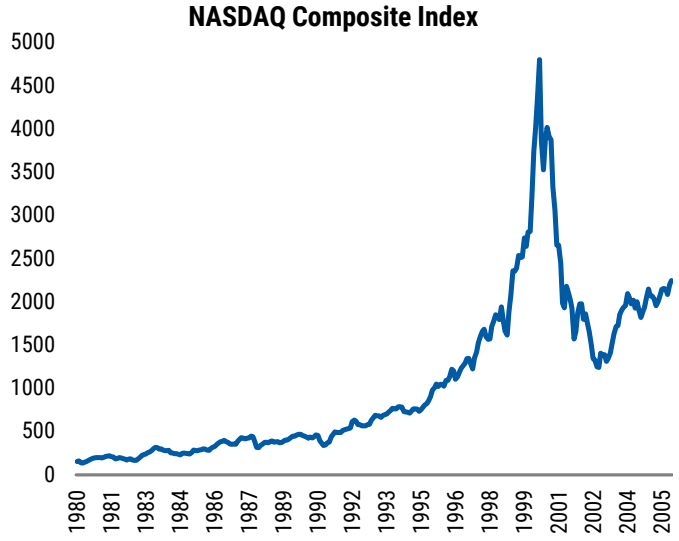
As was the case in earlier innovation waves, innovation and heavy capital spending led to boom-bust cycles. Prior overinvestment cycles were related to unprofitable investments infrastructure and other physical capital and the dot-com bubble mimicked these characteristics in intangible capital. The era saw speculative enthusiasm surrounding transformative technologies.

Equity valuations of internet firms rose rapidly before collapsing. However, the underlying infrastructure comprised of fiber networks, data centers, and software systems, provided the foundation for subsequent recovery. Venture capital funding surged again in the 2010s around social media, cloud computing, and artificial intelligence, reflecting recurring cycles of exuberance and consolidation.

Low interest rates and abundant liquidity likely amplified valuations, while expectations of network dominance drove capital toward a small set of firms. A key conclusion from our perspective is that financial markets both accelerate diffusion by funding risky ventures and amplify volatility when expectations adjust. Future Federal Reserve Chair Ben Bernanke, along with Mark Gertler and Simon Gilchrist would later win the Frontiers of Knowledge Award for their work on the financial accelerator, explaining how endogenous

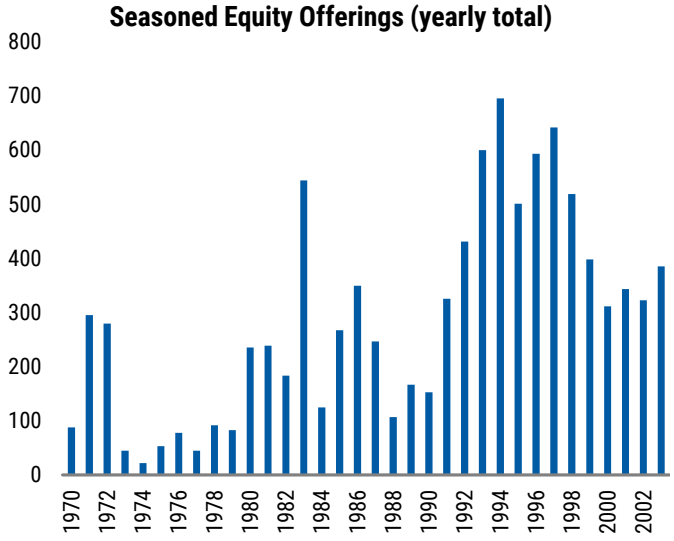
developments in credit markets work to amplify and propagate shocks to the macroeconomy. [3]

Exhibit 54: Exuberance about the technology boom sent NASDAQ stocks to record highs...



Source: Wall Street Journal, Haver Analytics, Morgan Stanley Research

Exhibit 55: ...and roughly tripled the number of seasoned equity offerings from 1990 onwards



Source: Ritter (2015). Morgan Stanley Research

3. Ben Bernanke, with Doublas Diamond and Phillip Dybvig, won their Nobel Prize for research that "significantly improved our understanding of the role of banks in the economy, particularly during financial crises". However, the work credited was largely related to the Great Depression.

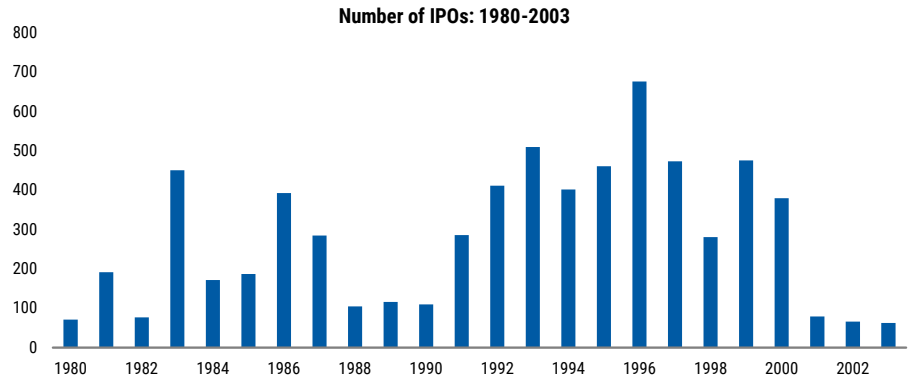
Exhibit 56:

Initial Public Offerings Surged during the tech boom, as did first-day returns

Year	Number of IPOs	Initial Public Offerings (IPOs) in the US Mean First-day Return (equal weighted)	Mean First-day Return (proceeds weighted)	Total Proceeds ($bn)
1980	71	14.3%	20.0%	0.9
1981	192	5.9%	5.7%	2.3
1982	77	11.0%	13.3%	1.0
1983	451	9.9%	9.4%	8.9
1984	172	3.6%	2.5%	2.1
1985	187	6.4%	5.3%	4.3
1986	393	6.1%	5.1%	13.4
1987	285	5.6%	5.7%	11.7
1988	105	5.5%	3.4%	3.9
1989	116	8.0%	4.7%	5.8
1990	110	10.8%	8.1%	4.3
1991	286	11.9%	9.7%	15.4
1992	412	10.3%	8.0%	22.7
1993	510	12.7%	11.2%	31.4
1994	402	9.6%	8.3%	17.2
1995	461	21.2%	17.5%	28.0
1996	677	17.2%	16.1%	42.1
1997	474	14.0%	14.4%	31.8
1998	281	21.9%	15.6%	33.7
1999	476	71.2%	57.4%	64.7
2000	380	56.4%	46.0%	64.8
2001	79	14.2%	8.7%	34.2
2002	66	9.1%	5.1%	22.0
2003	63	11.7%	10.4%	9.5
1980-1989	2049	7.2%	6.1%	54.2
1990-1998	3613	14.8%	13.3%	226.4
1999-2000	856	64.6%	51.7%	129.5

Source: Ritter, Jay R., Qie (Ellie) Yin, and Hongyu (Derek) Shan, 2017, Initial Public Offerings: Updated Statistics. Report No. 352.846-2837. University of Florida. Feb. 20, 2017. Morgan Stanley Research

Exhibit 57: Initial Public Offerings in the United States



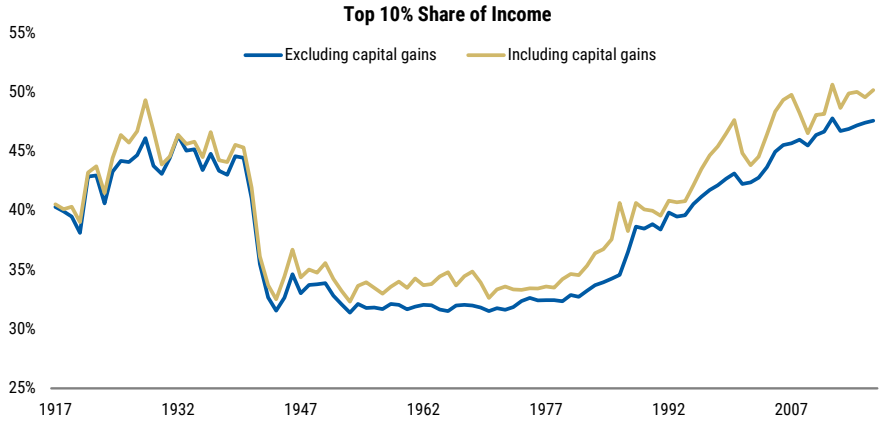
Source: Ritter, Jay R., Qie (Ellie) Yin, and Hongyu (Derek) Shan, 2017, Initial Public Offerings: Updated Statistics. Report No. 352.846-2837. University of Florida. Feb. 20, 2017. Morgan Stanley Research

Inequality: Income and Wealth Concentration

Whereas prior innovation waves were characterized by growing (or diminishing) income inequality, the internet and digital revolution saw both wider income and wealth inequality. Since the 1980s, top income and wealth shares have risen markedly, with the top decile of income earning household holding the vast majority of net financial wealth in the economy (Piketty and Saez 2003; Saez and Zucman 2016). Digital technologies have amplified returns to capital, superstar firms, and highly educated labor, contributing to concentration in both product and labor markets (Autor et al. 2020).

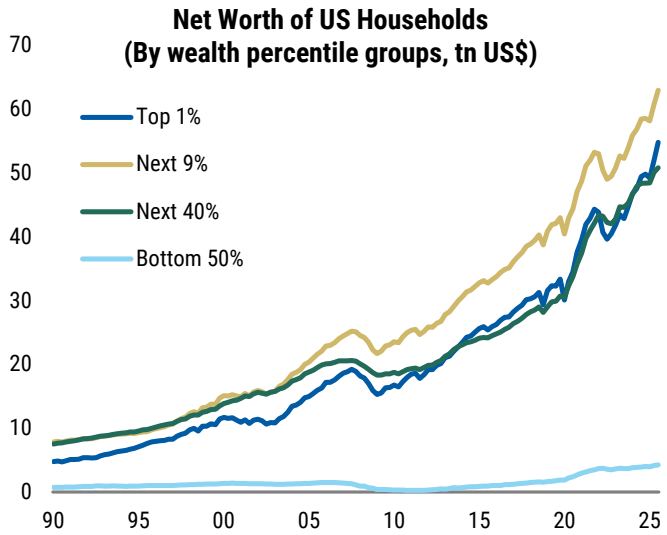
As the US economy has transformed away from agriculture to heavy industry and then to knowledge-based digital technologies, returns to capital have increased while returns to labor have diminished. Some studies point to equity compensation and capital gains tied to technology firms as having disproportionately benefited top earners, while routine wage growth has lagged. The post-1980 institutional environment has also featured declining unionization, lower marginal tax rates, financial globalization, and fewer labor protections. Wealth accumulation increasingly reflects ownership of intangible and financial assets rather than physical capital alone.

Exhibit 58: Income inequality widened after 1980 due, in part, to higher returns to skilled labor, higher returns to capital, and lower marginal tax rates



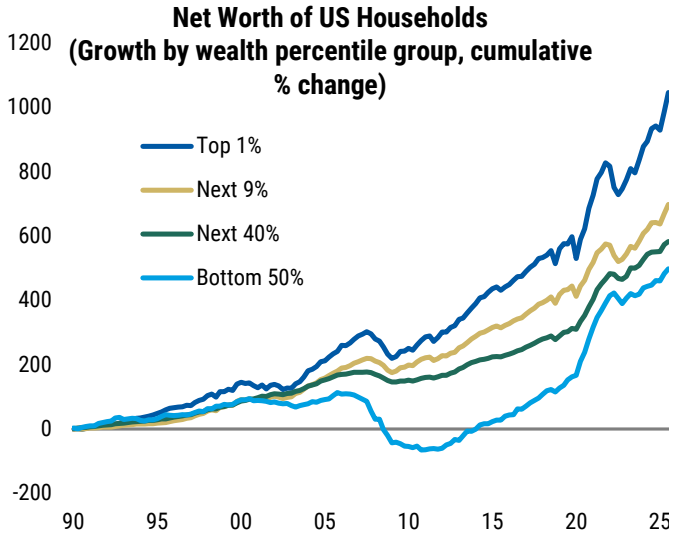
Source: Saez, Emmanuel, 2017."Striking it Richer: The Evolution of Top Incomes in the United States, Updated with 2017 final estimates," UC Berkeley. Morgan Stanley Research

Exhibit 59: Wealth inequality also widened, with the top 10% of households...



Source: Federal Reserve, Haver Analytics, Morgan Stanley Research

Exhibit 60: ...controlling about half of all household net worth



Source: Federal Reserve, Haver Analytics, Morgan Stanley Research

Politics and Social Response: From optimism to social division

Public debate has shifted in the direction of antitrust enforcement, data privacy, algorithmic bias, labor market disruption, and national competitiveness in advanced technologies. Public concern about misinformation, platform governance, and the influence of large technology firms on democratic processes have intensified. These have become features of the US public debate and the debate in other western democracies.

Governments, generally speaking, have responded with regulatory initiatives targeting digital competition and cross-border data flows. As in earlier innovation waves, rapid structural change has generated both optimism about productivity gains and concern about inequality, job security, and social cohesion. In the 1990s and early 2000s, many observers argued that the internet would decentralize information, lower entry barriers to speech, and strengthen democratic participation by empowering individuals relative to gatekeepers (Benkler 2006; Shirky 2008).

More recently, however, research has highlighted countervailing forces like algorithmic bias, network effects, and platform concentration that have often amplified misinformation and political polarization (Allcott and Gentzkow 2017; Sunstein 2018). At the same time, academic studies point to rising social isolation and declines in civic engagement that may be associated with digital substitution for face-to-face interaction (Putnam 2000; Twenge 2017). The policy challenge mirrors earlier eras: how to balance innovation incentives with safeguards against concentration, instability, and erosion of social capital.

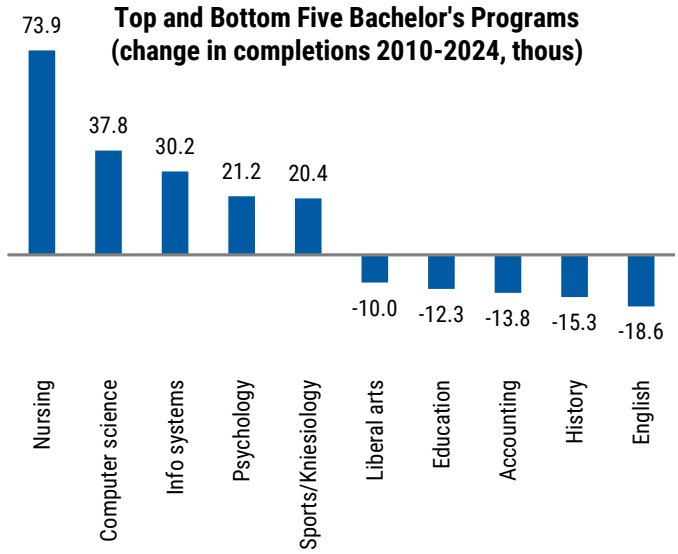
Education: From a focus on STEM to is college worth it?

The digital economy increased the premium to higher education and STEM skills, intensifying the long-running "race between education and technology" (Goldin and Katz 2008). Demand has grown not only for advanced technical expertise but also for complementary skills in problem-solving, communication, and adaptability.

Many universities and community colleges shifted their course and degree offerings toward computer science, data analytics, and engineering programs, while online learning platforms have broadened access to technical training. Yet rising tuition costs, unequal K–12 preparation, and geographic disparities in school quality risk widening skill gaps. Today, a growing number of high school graduates doubt whether a college degree is worth the cost involved. This development is a stark change from the way higher education has been viewed for the better part of a century.

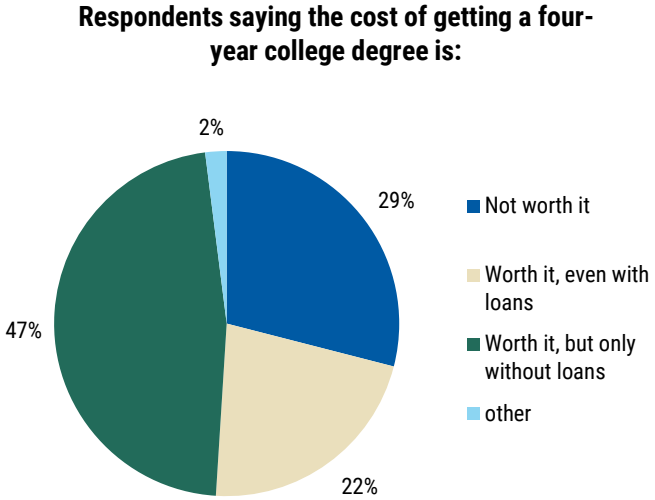
The fifth innovation wave may have shifted the balance toward lifelong learning and workforce retraining, as rapid technological change shortens the effective shelf life of many occupational skills and increases the need for continuous human capital investment.

Exhibit 61: STEM majors continue to grow in numbers...



Source: EAB, Morgan Stanley Research

Exhibit 62: ...though students are increasingly questioning the cost of an education



Source: Pew Research Center, Morgan Stanley Research

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